

A high-resolution analysis of arrestin2 interactions responsible for CCR5 endocytosis

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eLife Assessment

The authors investigate arrestin2-mediated CCR5 endocytosis in the context of clathrin and AP2 contributions. Using an extensive set of NMR experiments, and supported by microscopy and other biophysical assays, the authors provide **compelling** data on the roles of AP2 and clathrin in CCR5 endocytosis. This **important** work will appeal to an audience beyond those studying chemokine receptors, including those studying GPCR regulation and trafficking. The distinct role of AP2 and not clathrin will be of particular interest to those studying GPCR internalization mechanisms.

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Abstract

Clathrin-mediated endocytosis (CME) is crucial for regulating G protein-coupled receptors (GPCRs) via phosphorylation-dependent arrestin interactions. Despite detailed structural knowledge on the arrestin interactions with phosphorylated tails of GPCRs, the interplay between receptor phosphorylation and arrestin coupling to the CME machinery is not well understood, in particular due to the weakness and dynamics of the individual molecular interactions. Here we have characterized the interactions of arrestin2, which is activated by the phosphorylated C-terminus of the human chemokine receptor 5 (CCR5), with the main protein constituents of CME, namely clathrin and AP2 by solution NMR spectroscopy, biochemical and cellular assays. The NMR analysis revealed that arrestin2 interacts weakly with clathrin through a single binding site, independent of arrestin2 activation. In contrast, the arrestin2-AP2 interaction is stronger, requires arrestin2 activation by the CCR5 phospho-tail, and depends quantitatively on its degree of phosphorylation. These *in vitro* results are corroborated by cellular assays, which show that the chemokine-induced formation of a long-lived CCR5-arrestin2 internalization complex depends strongly on the interaction of arrestin2 with AP2, but not with clathrin. Taken together, these findings provide quantitative, atom-scale insights on the first steps of CCR5 endocytosis.

Introduction

G protein-coupled receptors (GPCRs) represent a large family of transmembrane cell surface receptors that mediate the cellular response to a wide variety of signals. Their crucial role in many human physiological processes makes them an important drug target with about a third of existing small-molecule drugs binding to GPCRs and an enormous potential for further development (Congreve et al., 2020). To ensure precise signal transduction the receptor signaling cascade events must be tightly regulated. This is achieved by coordinating receptor interactions with various intracellular proteins, including G proteins (Weis and Kobilka, 2018), GPCR kinases (GRKs) (Komolov and Benovic, 2018), and arrestins (Gurevich and Gurevich, 2019), with the latter eventually leading to receptor internalization in a process known as endocytosis. Endocytosis is a major mechanism of GPCR signal downregulation (Hanyaloglu and Zastrow, 2008). However, some receptors rapidly recycle back to the plasma membrane (Hanyaloglu and Zastrow, 2008), which may also serve to scavenge external ligands and degrade them within the lysosomes (Gu et al., 2023; Otero et al., 2006).

Although GPCR endocytosis can occur through several different pathways, the vast majority of GPCRs are internalized by clathrin-mediated endocytosis (CME) (Wolfe and Trejo, 2007). This process relies on the clathrin protein for the formation of vesicular structures from the plasma membrane that later mature into endosomes (Kaksonen and Roux, 2018; Smith and Smith, 2022). While clathrin exhibits a diverse interaction spectrum (Dell'Angelica, 2001; Lemmon and Traub, 2012), it cannot directly interact with the membrane receptors or plasma membrane lipids and requires adaptor proteins to select its internalization cargo (Kovtun et al., 2020; Popova et al., 2013; Smith et al., 1998; Wolfe and Trejo, 2007). The majority of GPCRs require arrestins as endocytic adaptors for internalization (Goodman et al., 1996). The interaction of GPCRs with arrestins is triggered by the phosphorylation of the GPCR C-terminal tail or intracellular loops by GRKs (Gurevich and Gurevich, 2019). Only two non-visual arrestins, arrestin2 and arrestin3, bind to hundreds of non-visual GPCRs (Gurevich and Gurevich, 2019). This is then followed by the interaction with the endocytic machinery (Moo et al., 2021), in particular clathrin (Goodman et al., 1997; Krupnick et al., 1997) and the adaptor complex 2 protein (AP2), which plays a central role in the formation of the clathrin-coated vesicles (Kelly et al., 2014) and the selection of the cargo molecules (Schmid et al., 2006).

We have recently investigated the phosphorylation and arrestin2 interactions of the human chemokine receptor 5 (CCR5) (Isaikina et al., 2023). CCR5 is a GPCR, which acts as the main HIV co-receptor (Alkhatib, 2009) and plays a central role in inflammation (Zeng et al., 2022), COVID-19 infection (Patterson et al., 2021), chronic diseases (Zeng et al., 2022), as well as cancer (Jiao et al., 2019). This revealed a key pXpp GPCR phosphorylation motif in the CCR5 C-terminal tail, which is specifically recognized by well-defined interactions of individual phosphates with positively charged arrestin2 residues. The pXpp motif is widely observed in other receptors (Isaikina et al., 2023; Maharana et al., 2023) known to form a stable complex with arrestin2 (Bous et al., 2022; Huang et al., 2020; Kang et al., 2015). The strength of the arrestin interaction correlates with the phosphosite density and its distribution on the receptor C-terminal tail and intracellular loops (Isaikina et al., 2023; Oakley et al., 2001).

While our previous work revealed the atomic details of the interactions between arrestin and multiple GPCR phosphosites leading to arrestin activation, the interplay between the activated arrestin with clathrin and AP2, which causes receptor internalization, remains poorly understood. Here we have characterized the interactions of arrestin2 with these two proteins as a function of arrestin2 activation by the phosphorylated CCR5 C-terminal tail using NMR and biochemical assays. The results establish strong activation-dependent interactions of arrestin2 with AP2, but

not with clathrin. These *in vitro* results are validated by cellular assays, which also demonstrate that CCR5 forms long-lived endocytosed complexes with arrestin2 upon agonist chemokine stimulation.

Results

Localizing and quantifying the arrestin2-clathrin interaction by NMR spectroscopy

The CCR5 receptor relies on arrestin2/3 for internalization (Fraile-Ramos et al., 2003) via clathrin-mediated pathways (Signoret et al., 2005). Arrestin2 consists of two consecutive (N- and C-terminal) β -sandwich domains (Figure 1A), followed by the disordered clathrin-binding loop (CBL, residues 353–386), strand β 20 (residues 386–390), and a disordered C-terminal tail after residue 393. The CBL contains a clathrin-binding motif ([L/I] Φ X Φ [D/E] where X denotes any amino acid and Φ is a bulky hydrophobic residue (Dell’Angelica, 2001; Smith et al., 2017) (Figure 1A, residues 376–380, Figure 1-figure supplement 1A). Strand β 20 forms a characteristic β -sheet with the N-terminal strand β 1 in inactive arrestin. Upon activation, the phosphorylated receptor tail replaces β 20 in the β -sheet and releases it from the arrestin core.

Clathrin and arrestin interact in their basal state (Goodman et al., 1996), and a structure of a complex between arrestin2 and the clathrin heavy chain N-terminal domain (residues 1–363, named clathrin-N in the following) has been solved by X-ray crystallography in the absence of an arrestin2-activating phosphopeptide (PDB:3GD1) (Kang et al., 2009). This structure (Figure 1-figure supplement 1B) suggests a 2:1 binding model between arrestin2 and clathrin-N. The first interaction (site I) is observed between the ³⁷⁶LIELD³⁸⁰ clathrin-binding motif of the arrestin2 CBL and the edge of the first two β -sheet blades of clathrin-N, whereas the second interaction (site II) occurs between arrestin2 residues ³³⁴LLGDLA³³⁹ and the 4th and 5th blade of clathrin-N. The latter arrestin interaction site is not present in the arrestin2 splice variant arrestin2S (for short) where an 8-amino acid insert (residues 334–341) between β -strands 18 and 19 is removed (Kang et al., 2009).

To elucidate the clathrin-arrestin interactions under solution conditions and the influence of arrestin activation we utilized NMR spectroscopy. Figure 1B depicts part of a ¹H-¹⁵N TROSY spectrum (full spectrum in Figure 1-figure supplement 2A) of the truncated ¹⁵N-labeled arrestin2 construct arrestin2^{1–393} (residues 1–393), which encompasses the C-terminal strand β 20, but lacks the disordered C-terminal tail. Due to intrinsic microsecond dynamics, the assignment of the arrestin2^{1–393} ¹H-¹⁵N resonances by triple resonance methods is largely incomplete, but 16 residues (residues 367–381, 385–386) within the mobile CBL could be assigned. This region of arrestin is typically not visible in either crystal or cryo-EM structures due to its high flexibility. The addition of an equimolar amount of clathrin-N (clathrin heavy chain 1, residues 1–364) to arrestin2^{1–393} induced significant line broadening from residue T374 to T381 with I377 being most strongly affected (Figure 1C, Figure 1-figure supplement 2B). This region almost exactly matches and fully contains the canonical arrestin2 clathrin-binding motif ³⁷⁶LIELD³⁸⁰. Figure 1A shows these interacting residues mapped on an arrestin2 AlphaFold2 model (Jumper et al., 2021).

As ¹H-¹³C methyl resonances have more favorable relaxation properties than backbone ¹H-¹⁵N resonances (Tugarinov et al., 2003), we used Ile- δ 1-¹³CH₃-labeled arrestin2 to quantitatively monitor the formation of the arrestin2-clathrin complex (arrestin2 assignments transferred from BMRB, ID:51131 (Shiraishi et al., 2021)). HMQC spectra of the 15 Ile δ 1-¹³CH₃ methyl groups of arrestin2^{1–393} (Figure 1D) revealed changes in peak intensity (Figure 1E, top) and chemical shift perturbations (Figure 1E, bottom) only for I377 at an equimolar addition of unlabeled clathrin-N to arrestin2^{1–393}. An apparent dissociation constant $K_D = 104 \pm 9 \mu\text{M}$ was determined

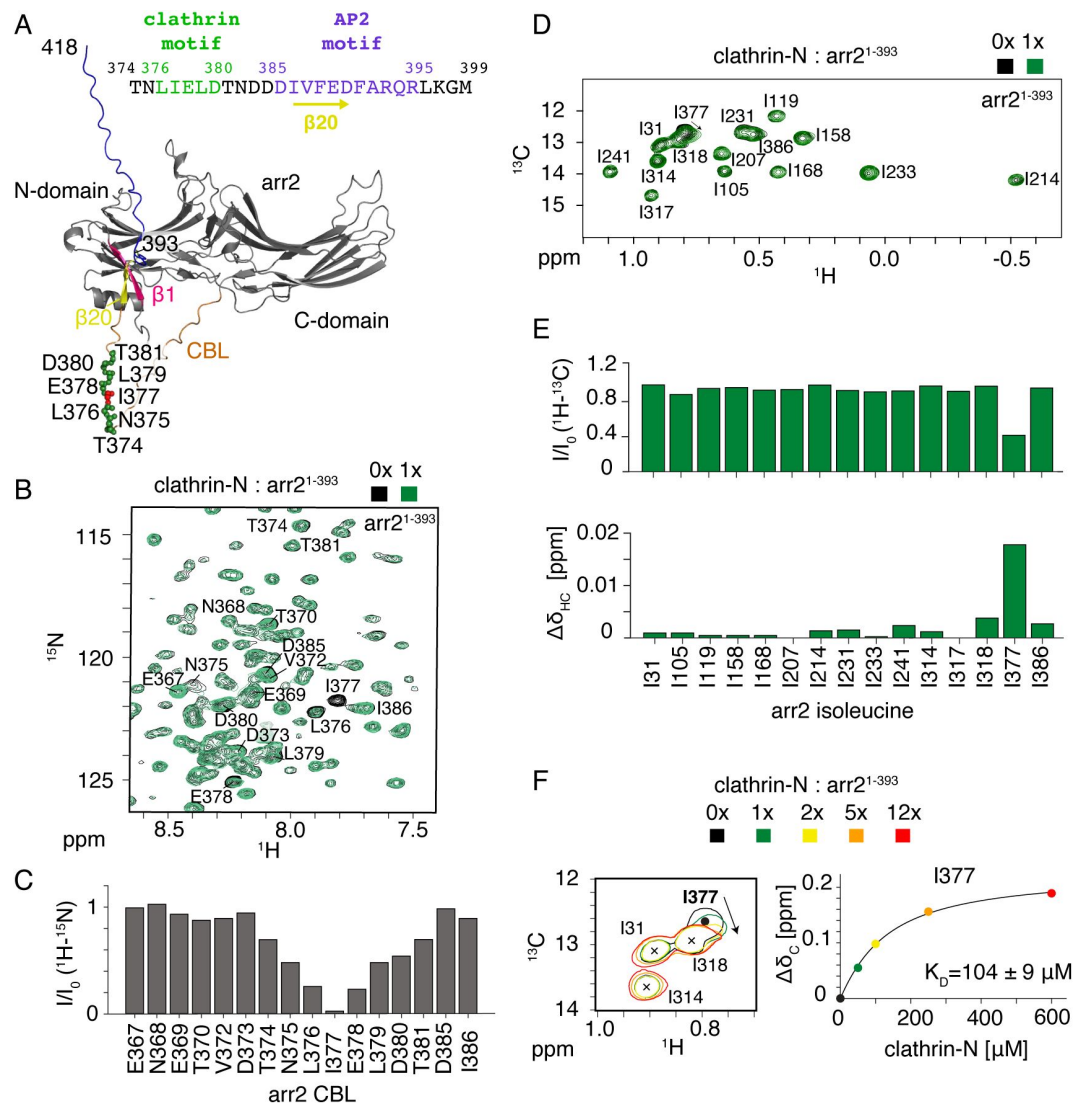


Figure 1

Arrestin2-clathrin-N interactions detected by arrestin2 NMR spectra.

(A) AlphaFold2 model of arrestin2¹⁻⁴¹⁸ (arr2) showing the N- and C-domain (gray) followed by the clathrin-binding loop (CBL, orange), β 20 strand (yellow) and the disordered C-terminal tail (blue). The model was used to visualize the clathrin-binding loop and the 344-loop of the arrestin2 C-domain, which are not detected in the available crystal structures of apo arrestin2 [bovine: PDB 1G4M (Han et al., 2001), human: PDB 8AS4 (Isaikina et al., 2023)]. In the other structured regions, the model is virtually identical to the crystal structures. The C-terminal strand β 20 forms a parallel β -sheet with the N-terminal strand β 1 (pink) in inactive arrestin2 and is released from arrestin2 upon activation by phosphopeptides. Arrestin2 residues interacting with clathrin-N are shown as green (red for main interacting residue I377) spheres. The clathrin and AP2 binding motifs are indicated in the arrestin2 sequence on the top. (B) Part of ^1H - ^{15}N TROSY spectrum of apo arrestin2¹⁻³⁹³ (black) and upon addition of an equimolar amount of clathrin-N (green). Assigned CBL arrestin2¹⁻³⁹³ residues are indicated. (C) Intensity ratios of assigned complexed vs. apo arrestin2¹⁻³⁹³ CBL resonances. The region between T374 and T381 undergoes significant intensity attenuation. (D) ^1H - ^{13}C HMQC spectra of apo Ile- δ - ^{13}C ^{13}H -labeled arrestin2¹⁻³⁹³ (black) and upon addition of an equimolar amount of clathrin-N (green). (E) Intensity attenuation (top) and chemical shift perturbation (bottom) of arrestin2¹⁻³⁹³ isoleucine ^1H - ^{13}C $^{\delta 1}$ resonances upon clathrin-N addition. Out of 15 isoleucine residues, significant changes are observed only for I377. (F) Left: part of ^1H - ^{13}C HMQC spectrum showing resonance shifts of I377 upon clathrin-N binding. Right: detected chemical shift changes as a function of clathrin-N concentration. The solid line depicts a non-linear least-square fit to the data points with respective dissociation constant.

from titrating clathrin-N and fitting the arrestin2 I377 ^{13}C chemical shifts with a 1:1 binding isotherm (**Figure 1F**). Although the formed complex (total molecular weight 85 kDa) has low affinity, such interactions are common in endocytosis, which involves many multi-site interactions within a large network of proteins (Smith et al., 2017).

The arrestin2-clathrin interaction is independent of arrestin2 activation

Due to significant line broadening and peak overlap of the arrestin2 resonances upon phosphopeptide addition, the influence of arrestin activation on the clathrin interaction could not be detected on either backbone or methyl resonances. However, clathrin-N, although similar in size to arrestin2¹⁻³⁹³ (41 kDa vs. 44 kDa), provided considerably better ^1H - ^{15}N TROSY spectra due to the absence of significant micro- to millisecond dynamics. This allowed to observe the effect of arrestin2 activation on its interaction with clathrin by titrating arrestin2¹⁻³⁹³ to ^{15}N -labeled clathrin-N.

Figure 2A (left) shows the intensity changes (full spectra in **Figure 2-figure supplement 1A**) of the clathrin-N ^1H - ^{15}N TROSY resonances [assignments transferred from BMRB, ID:25403 (Zhuo et al., 2015)] upon addition of a one-molar equivalent of arrestin2¹⁻³⁹³. A significant intensity reduction due to line broadening is detected for clathrin-N residues 39-40, 48-50, 62-72, 83-90, 101-106, and 108. These residues form a clearly defined binding region at the edges of blade 1 and blade 2 of clathrin-N (**Figure 2A**, right), which corresponds to interaction site I in the 3GD1 crystal structure, involving the conserved arrestin2 ³⁷⁶LIED³⁸⁰ motif. However, no significant signal attenuation was observed for clathrin-N residues in blade 4 and blade 5, which would correspond to the crystal interaction site II with arrestin2 residues ³³⁴LLGDLA³³⁹ that are absent in the arrestin2S splice variant. Thus, only one arrestin2 binding site in clathrin-N is detected in solution, and site II of the crystal structure may be a result of crystal packing.

Next, we tested whether the arrestin2-clathrin-N interaction is enhanced in the presence of the fully phosphorylated CCR5 phosphopeptide (CCR5pp6, where 6 indicates the number of phosphosites, **Table S1**), which displaces arrestin2 strand β 20 by forming an antiparallel β -sheet with strand β 1, thereby activating arrestin2 (Isaikina et al., 2023). However, no additional signal attenuation of the clathrin-N ^1H - ^{15}N resonances or extension to further clathrin-N regions was observed (**Figure 2B**, **Figure 2-figure supplement 1B**), when arrestin2¹⁻³⁹³ was added in the presence of a 5-molar equivalent of CCR5pp6 (corresponding to 80% of arrestin activation). Thus, activation of arrestin2 has apparently no effect on its interaction with clathrin-N. These results suggest that even for receptors with low phosphosite density, such as adrenergic and dopamine receptors (Isaikina et al., 2023; Oakley et al., 2001), the arrestin2-clathrin interaction may still take place allowing efficient receptor internalization.

Previous studies have suggested that the last residues of the arrestin2 C-terminus have an inhibitory effect on clathrin binding (Kern et al., 2009). We therefore tested the interaction of ^{15}N -labeled clathrin-N with full-length arrestin2 (arrestin2¹⁻⁴¹⁸). However, the obtained residue intensity attenuation was indistinguishable from the one observed for the arrestin2¹⁻³⁹³ construct (**Figure 2C**, **Figure 2-figure supplement 1C**). Therefore, the residues of the arrestin2 C-terminal tail do not interfere with clathrin binding. In contrast, the interaction of clathrin with arrestin2 (**Figure 2D**, **Figure 2-figure supplement 1D**) was completely abolished when the arrestin2 clathrin-binding loop was replaced by the loop of visual arrestin (arrestin2^{ACBL}, construct details in Methods). The latter loop lacks the clathrin-binding motif making visual arrestin incapable of interacting with clathrin (Goodman et al., 1997).

Taken together, the solution NMR experiments reveal that arrestin2 interacts with clathrin-N through a single binding site, independently of arrestin2 activation and the conformation of the arrestin2 core.

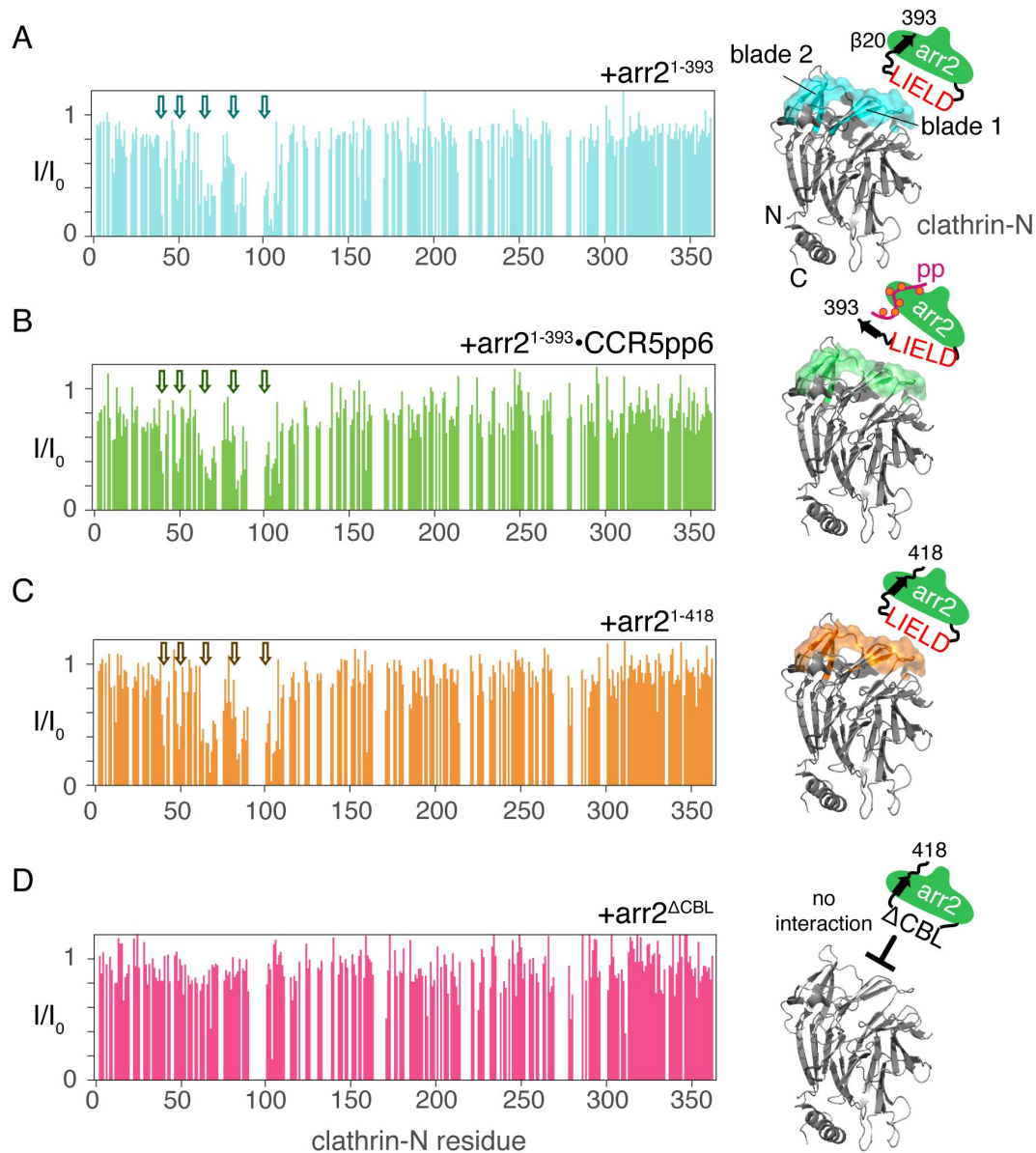


Figure 2

Arrestin2-clathrin-N interactions detected by clathrin-N NMR spectra.

(Left): intensity attenuation I/I_0 of clathrin-N ^1H - ^{15}N TROSY resonances upon addition of equimolar amounts of various arrestin2 constructs: (A) apo arrestin2¹⁻³⁹³, (B) CCR5pp6-activated arrestin2¹⁻³⁹³, (C) full-length arrestin2¹⁻⁴¹⁸ and (D) arrestin2^{ΔCBL} with removed clathrin-binding motif. Regions that undergo significant attenuation upon interaction with arrestin2 are indicated by arrows. Right: residues undergoing significant intensity attenuation (one standard deviation below the average signal attenuation) are marked on the structure of clathrin-N (PDB: 3GD1) together with a schematic representation of the respective arrestin2 construct.

Arrestin activation is essential for AP2 interaction

We then proceeded to characterize the interaction of arrestin2 with AP2. AP2 consists of two large (α and β 2) and two smaller (μ 2 and σ 2) subunits (Collins et al., 2002). While each subunit plays a role in CME (Smith et al., 2017), direct recognition of endocytic cargo motifs has been attributed to the σ 2 subunit (LL-motif) and μ 2 subunit (Yxx Φ -motif) (Jackson et al., 2010). The β 2 subunit of the AP2 complex (AP2 β 2) interacts with the arrestin2/3 C-terminus (Kim and Benovic, 2002; Laporte et al., 2002, 2000), and a crystal structure of the complex between the AP2 β 2 C-terminal domain and a C-terminal arrestin2 peptide (residues 383–402) encompassing strand β 20 has been solved (PDB: 2IV8) (Schmid et al., 2006). Interestingly, the C-terminal arrestin2 peptide forms a helical structure in the complex with AP2 (Figure 3A) as opposed to its β -sheet coordination with strand β 1 in inactive arrestin2. No experimental data exist on the formation of the AP2-arrestin2 complex in the context of full-length arrestin2 and its activation by phosphorylated receptor tails.

To test how different phosphorylation levels influence arrestin2 recruitment by the C-terminal domain of AP2 β 2 (AP2 β 2^{701–937}, residues 701–937), we performed a SEC binding assay. In the absence of phosphopeptide an equimolar mixture of 10 μ M arrestin2^{1–418} and 10 μ M AP2 β 2 is separated into two peaks (Figure 3B, orange), which elute at the positions corresponding to the individual apo proteins (Figure 3B, blue, green). Upon addition of 100 μ M CCR5pp6 (Figure 3C, purple), an additional SEC peak appears at an elution volume of 2.6 ml, which corresponds to the formed arrestin2-AP2 β 2 complex. Monotonously decreasing complex yields were observed when reducing the amounts of CCR5pp6 added to the mixture (Figure 3D), and no complex formation occurred in the presence of 100 μ M non-phosphorylated CCR5 phosphopeptide (CCR5pp0) (Figure 3E). The dissociation constant of the CCR5pp6-arrestin2 complex was determined previously as 45 μ M (Table S1) (Isaikina et al., 2023). Not surprisingly, the formation of the arrestin2-AP2 β 2 complex depends quantitatively on the affinity of the phosphopeptide-arrestin2 complex. Correspondingly, the addition of the less phosphorylated CCR5pp3 and CCR5pp4 with lower affinities towards arrestin2 (both $K_D \sim 200$ μ M, Table S1) (Isaikina et al., 2023), resulted in a considerable, similar reduction of the formed arrestin2-AP2 β 2 complex (Figure 4F).

Using the extinction coefficients of all involved polypeptides, the apparent affinities between AP2 β 2 and arrestin2 could be determined from the SEC profiles (Figure 3G, see Materials and Methods for details). In the presence of 100 μ M CCR5pp6, this apparent K_D amounts to 9 μ M. An identical affinity value for the affinity between AP2 β 2 and an isolated arrestin β 20 peptide (residues 383–402, D402C) has been observed by EPR spectroscopy (Moaven et al., 2013). In the presence (100 μ M) of the less phosphorylated CCR5 peptides, the apparent AP2 β 2-arrestin2 affinities are reduced by ~ 4 –5 times (CCR5pp4: 44 μ M, CCR5pp3: 40 μ M), in good agreement with their ~ 4 –5 times reduced affinity for arrestin2.

We also attempted to stabilize the arrestin2-AP2 β 2-phosphopeptide complex through the addition of PIP2, which can stabilize arrestin complexes with the receptor (Janetzko et al., 2022). The addition of PIP2 increased the formation of arrestin2 dimers and higher oligomers, presumably due to the presence of additional charges. Unfortunately, the resolution of the SEC experiments was not sufficient to separate the arrestin2 oligomers from complexes with AP2 β 2.

It is interesting to note that CCR5pp3 and CCR5pp4 have similar affinities towards arrestin2 despite their different phosphorylation levels (3-fold vs. 4-fold phosphorylated), but only CCR5pp3 contains the pXpp barcode which increases the arrestin2 activation, as assayed by the binding of the Fab30 antibody (Isaikina et al., 2023). Apparently, this barcode does not play a significant role in the formation of arrestin2-AP2 β 2 complex as the complex formation appears solely governed by the affinity of the phosphopeptide towards arrestin2.

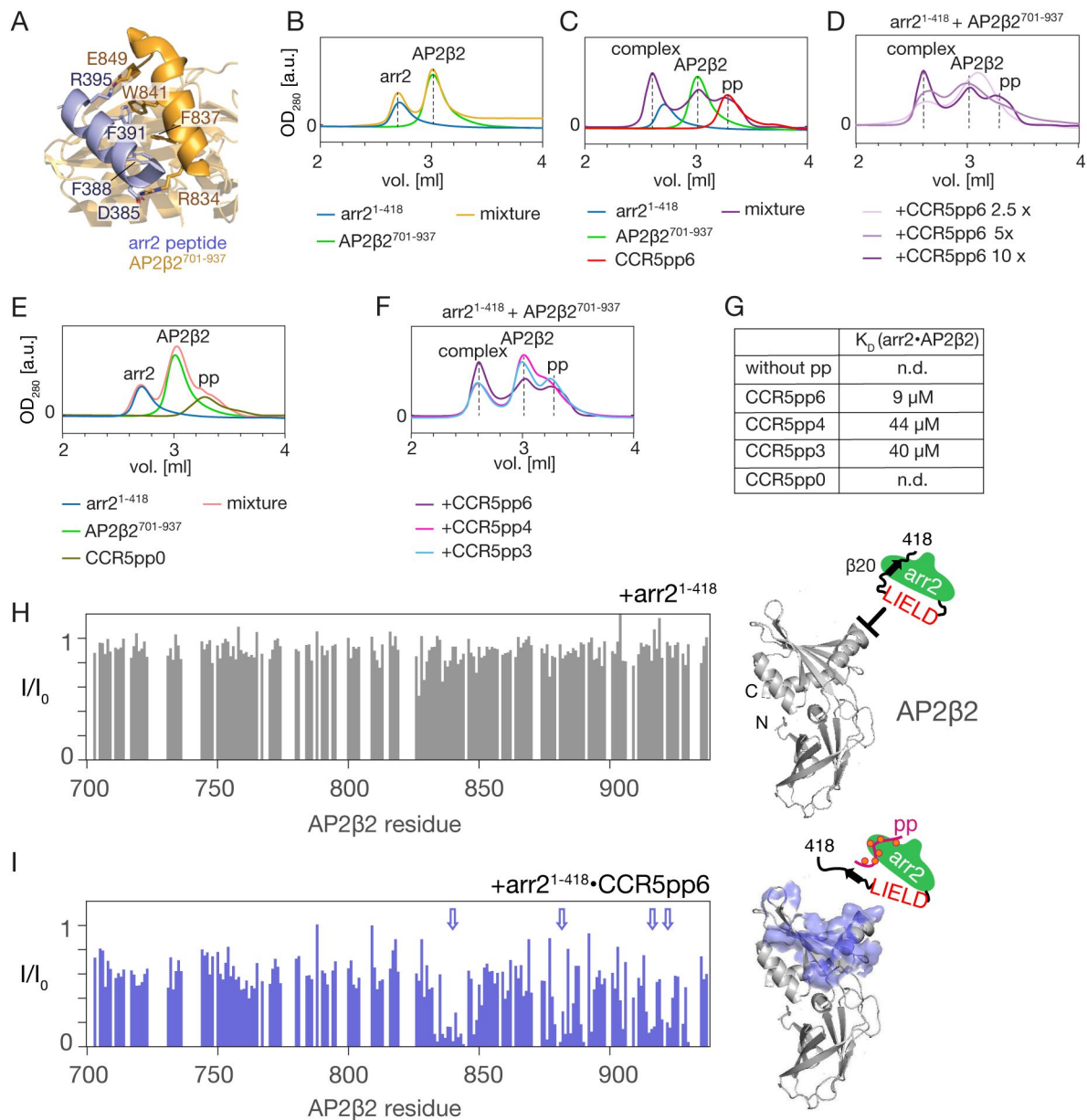


Figure 3

Arrestin2 interaction with the C-terminal domain of the AP2β2.

(A) Structure of AP2β2⁷⁰¹⁻⁹³⁷ (orange) in complex with arrestin C-terminal peptide (blue) (PDB: 2IV8). Important residues from both chains stabilizing the interaction are depicted in stick representation. (B-F) SEC profiles of arrestin2¹⁻⁴¹⁸, AP2β2⁷⁰¹⁻⁹³⁷, various phosphopeptides and their mixtures. The annotated color coding below the SEC profile indicates the individual sample composition. In panels (B, C, E) 'mixture' refers to a SEC sample containing all of the individual components indicated on the left. In panels (D, F) the primary sample composition is indicated above the SEC profile, and the annotated color coding below the SEC profile indicates the added component to the primary sample composition. (G) Apparent affinities for arrestin2-AP2 complex formation in the presence of phosphopeptides derived by integrating the SEC peaks marked in **Figure 4F** and scaling the integrals by the respective extinction coefficients. 'nd' indicates 'not detectable'. (H, I) (Left): intensity attenuation I/I_0 of AP2β2⁷⁰¹⁻⁹³⁷ ¹H-¹⁵N TROSY resonances upon addition of equimolar amounts of apo arrestin2¹⁻⁴¹⁸ (H) or CCR5pp6-activated arrestin2¹⁻⁴¹⁸ (I). Regions that undergo extensive specific attenuation upon interaction with arrestin2 are indicated by arrows. Right: residues undergoing extensive specific intensity attenuation are marked on the structure of AP2β2 (PDB: 2IV8) together with a schematic representation of the respective arrestin2 construct.

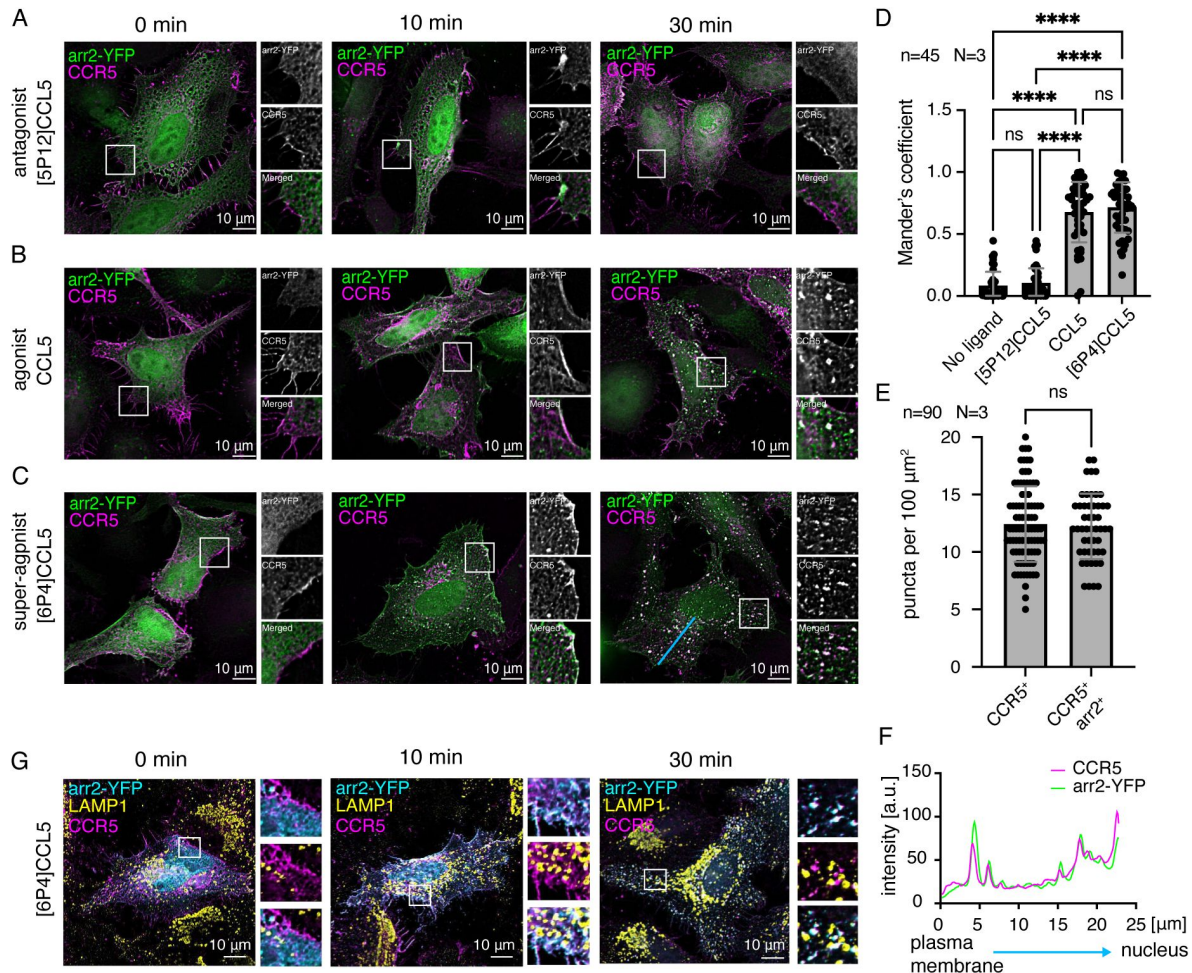


Figure 4

CCR5 internalization in the presence of the chemokine ligands.

CCR5 internalization monitored in HeLa cells co-transfected with plasmids containing arrestin2-YFP (green) and CCR5 genes (magenta) stimulated with (A) [5P12]CCL5 (antagonist), (B) CCL5 (natural agonist), (C) [6P4]CCL5 (super-agonist). Cells were stimulated for 0, 10, and 30 min with chemokine ligands before fixation and preparation for fluorescence microscopy. Each image is accompanied by a zoomed region of interest (white squares) showing deconvolved CCR5 and arrestin2-YFP signals. (D) Mander's colocalization coefficients of the CCR5 and arrestin2-YFP signals in the absence and presence of chemokine ligands 30 min after stimulation. Individual values, as well as mean and standard deviation are shown for N=3 biological replicates and n=45 ROIs from 20 cells; ****: $P < 0.0001$; ns: not significant ($P > 0.9999$). (E) Density of CCR5-positive (CCR5⁺) and CCR5/arrestin2-positive (CCR5⁺/arrestin2⁺) puncta after 30 min [6P4]CCL5 ligand stimulation. Individual values, as well as mean and standard deviation are shown for N=3 biological replicates and n=90 ROIs from 30 cells; ns: not significant ($P > 0.9999$). (F) CCR5 and arrestin2 fluorescence signals along the trajectory from the plasma membrane to the nucleus (blue line in panel c) 30 min after [6P4]CCL5 ligand stimulation. (G) Lysosomal trafficking of the CCR5 (magenta) and arrestin2 (cyan) complex in presence of [6P4]CCL5 monitored in the HeLa cells using LAMP1 antibodies (yellow). No recruitment to the lysosome is observed.

To observe the atomic details of arrestin2 activation on its interaction with AP2 β 2, we used ^{15}N -labeled AP2 β 2⁷⁰¹⁻⁹³⁷ [assignment transferred from BMRB ID:52320 (Naudi-Fabra et al., 2024)]. Upon addition of arrestin2¹⁻⁴¹⁸, no changes in the AP2 β 2 ^1H - ^{15}N spectra are detected (Figure 3H, Figure 3-figure supplement 1A), apparently due to the unavailability of arrestin2 β 20 strand and C-terminal tail. However, when arrestin2¹⁻⁴¹⁸ was added in the presence of a 5-molar equivalent of CCR5pp6 (corresponding to 80% of arrestin activation), a significant intensity reduction due to line broadening is detected for many AP2 β 2 ^1H - ^{15}N resonances (Figure 3I, left, Figure 3-figure supplement 1B). The strongest attenuated resonances clearly define a binding region within the C-terminal domain of AP2 β 2 (Figure 3I, right), which is an extension of the binding surface observed in the crystal structure (PDB: 2IV8) of AP2 β 2 in complex with a 21-residue arrestin2 C-terminal peptide containing strand β 20 (Schmid et al., 2006). In agreement with the AP2 β 2 NMR observations, no interaction was observed in the arrestin2 methyl and backbone NMR spectra upon addition of AP2 β 2 in the absence of phosphopeptide (Figure 3-figure supplement 1C, D). However, the significant line broadening of the arrestin2 resonances upon phosphopeptide addition (Figure 3-figure supplement 1E, F) precluded a meaningful assessment of the effect of the AP2 β 2 addition on arrestin2 in the presence of phosphopeptide. The observed line broadening of arrestin2 in the presence of phosphopeptide must be a result of increased protein motions and is not caused by a decrease in protein stability, since the melting temperature of arrestin2 in the absence and presence of phosphopeptide are identical (56.9 ± 0.1 °C).

We then tested whether the presence of clathrin-N influences complex formation between arrestin2 and AP2 β 2, and vice versa. Arrestin2¹⁻⁴¹⁸ does not form a complex in the SEC assay with clathrin-N (Figure 3-figure supplement 2A), which agrees with the weak affinity observed in the NMR experiments (Figure 1F). Furthermore, a SEC analysis of a mixture of arrestin2¹⁻⁴¹⁸, clathrin-N, AP2 β 2⁷⁰¹⁻⁹³⁷, and CCR5pp6 (Figure 3-figure supplement 2B) indicates formation of a complex at 2.6 ml elution volume. An SDS analysis of this complex fraction showed the presence of arrestin2¹⁻⁴¹⁸ and AP2 β 2⁷⁰¹⁻⁹³⁷, but not of clathrin-N (Figure 3-figure supplement 2C). This indicates that clathrin-N does not interact strongly with the arrestin2-AP2-CCR5 tail complex. As such clathrin may not engage with the initial receptor internalization complex, but may be recruited at later stages of the CME.

To test the stability of the formed arrestin2-AP2 interaction and its quaternary coordination, we performed a trypsin proteolysis assay (Figure 3-figure supplement 3, uncropped gels in Figure 3-figure supplement 4). Trypsin recognizes arrestin2 residue R393 (Xiao et al., 2004), which is buried in the arrestin2 apo state, but is involved in the interaction with AP2 β 2 (Laporte et al., 2000). Without phosphopeptide, the arrestin2¹⁻⁴¹⁸ digestion midpoint occurs at 20-30 min due to the inaccessibility of the cleavage site (Figure 3-figure supplement 3A-C). In the presence of a 5-molar equivalent of CCR5pp6 (Figure 3-figure supplement 3A), the top band corresponding to full-length arrestin2 has almost completely disappeared (relative abundance 19%) already 5 min after incubation as a consequence of the β 20 strand release and the exposure of R393 to trypsin. However, the presence of AP2 β 2⁷⁰¹⁻⁹³⁷ protects arrestin2¹⁻⁴¹⁸ from proteolysis, as evident from significantly higher abundance at 5 min and a shift of the proteolysis midpoints to 10, 20, and 30 min at 1-, 2-, and 5-molar equivalents of AP2 β 2⁷⁰¹⁻⁹³⁷ respectively.

Analogous digestion experiments using CCR5pp3 (Figure 3-figure supplement 3B) or CCR5pp4 (Figure 3-figure supplement 3C) phosphopeptides revealed that they did not enhance the rate of arrestin2 proteolysis by trypsin at a 3-molar equivalent, as expected from their weaker affinity towards arrestin2. In agreement with this, however, upon addition of 25-molar equivalents of either peptide the rate of arrestin2 proteolysis was significantly increased, as full-length arrestin2 almost completely disappeared already 5 min after incubation. Thus as for CCR5pp6, the presence of AP2 β 2⁷⁰¹⁻⁹³⁷ also protects arrestin2¹⁻⁴¹⁸ activated by CCR5pp3 or CCR5pp4 from proteolysis.

In summary, these results indicate that arrestin2 activation by the binding of receptor phosphopeptides and the subsequent release of its strand $\beta 20$ is required for the interaction with AP2. This interaction is modulated by the phosphorylation level of the receptor C-terminal peptide and its relative abundance. The formed AP2 complex with arrestin is stable and more resistant to proteolysis.

Ligand-dependent CCR5 internalization monitored in HeLa cells

We then validated the obtained *in vitro* results on the CCR5-arrestin2-clathrin and -AP2 interactions by cellular assays. According to the strength and persistence of arrestin interactions, GPCRs have been categorized into class A and class B subtypes (Isaikina et al., 2023 [↗](#); Oakley et al., 2000 [↗](#)). To distinguish these arrestin classes from the common IUPHAR classification of GPCRs, we refer in the following to them as arr-class A and B, respectively. Arr-class A receptors (e.g. adrenergic receptors) form transient complexes with arrestins that quickly dissociate near the plasma membrane and the receptor traffics alone into endosomes. In contrast, arr-class B receptors (e.g. the vasopressin receptor 2, V2R) have a higher affinity towards arrestin and form long-lived complexes that traffic together into endosomes (Oakley et al., 2000 [↗](#)). We have recently established that the phosphorylated CCR5 C-terminus forms a high-affinity complex with arrestin (Isaikina et al., 2023 [↗](#)), thus assigning the CCR5 to the arr-class B subtype.

To better understand the stability of CCR5-arrestin2 complex, we first monitored CCR5 internalization upon chemokine stimulation in fixed HeLa (CCL2, ATCC nomenclature) cells (**Figure 4** [↗](#)). HeLa cells were transiently transfected with plasmids containing C-terminally FLAG-tagged CCR5 and YFP-tagged arrestin2 (arrestin2-YFP) genes. This receptor construct was used to minimize CCR5 modifications and monitor the arrestin2 interaction in conditions as close to native as possible. Before the addition of the chemokine ligands (0 min), CCR5 was observed at the plasma membrane (magenta), while arrestin2 (green) was distributed throughout the cell (**Figure 4A-C** [↗](#)). Cells stimulated with the antagonist chemokine [5P12]CCL5 (Gaertner et al., 2008 [↗](#)) retained this CCR5 and arrestin2 distribution even 30 min after ligand addition (**Figure 4A** [↗](#)). In contrast, when cells were stimulated with the native agonist chemokine CCL5 (Hughes and Nibbs, 2018 [↗](#)) (**Figure 4B** [↗](#)) or the super-agonist [6P4]CCL5 (Gaertner et al., 2008 [↗](#); Isaikina et al., 2021 [↗](#)) (**Figure 4C** [↗](#)), recruitment of arrestin2 from the cytoplasm to the plasma membrane could be detected within 10 min. Prolonged ligand incubation times (30 min) lead to CCR5 internalization in complex with arrestin2. To eliminate the possibility that arrestin2 recruitment to the membrane was due to interactions with endogenous proteins in HeLa cells, we monitored internalization in live cells transfected only with arrestin2. In this case, the cells did not show characteristic arrestin2 recruitment to the plasma membrane or robust puncta formation, indicative of endosomes, even in the presence of the super-agonist (**Figure 4-figure supplement 1A** [↗](#)). This control experiment also revealed a few internal arrestin puncta devoid of receptor, presumably due to the arrestin2 overexpression. This phenomenon was also observed previously (Beautrait et al., 2017 [↗](#); Oakley et al., 2000 [↗](#)).

Stronger phosphorylation and arrestin2/3 recruitment potencies have been reported for [6P4]CCL5 as compared to wild-type CCL5 (Martins et al., 2020 [↗](#)). In agreement with this, [6P4]CCL5 appears more potent than CCL5 in recruiting arrestin2 to the plasma membrane and inducing CCR5-arrestin2 puncta formation after 10 min (**Fig 4B** [↗](#), C). However, 30 min after stimulation, no significant difference in CCR5 and arrestin2 colocalization was observed as quantified by the Mander's coefficient (**Figure 4D** [↗](#)). The Mander's coefficient is about 0.7 in cells stimulated by either agonist, but close to zero for non-stimulated or antagonist-incubated cells (**Figure 4D** [↗](#)). Apparently the two agonist ligands induce a differing initial rate of arrestin2 recruitment by CCR5. However, once arrestin2 is recruited, the CCR5-arrestin2 complex accumulates and remains stable for at least 30 min inside the cells.

To further corroborate the arr-class B-like behavior of CCR5, we counted the puncta positive for CCR5 or both CCR5 and arrestin2 in cells at this time point after stimulation by [6P4]CCL5 (**Figure 4E**). No significant difference was observed between the two values. This indicates that the number of puncta containing only CCR5, but not arrestin2, must be very small and that the majority of CCR5 puncta also contains arrestin2, consistent with stable complex formation. Furthermore, the CCR5 and arrestin2 colocalization occurs not only close to the plasma membrane but also close to the nucleus (**Figure 4F**), indicative of the arr-class B-like behavior of CCR5. For arr-class A receptors, it would be expected that arrestin2 is excluded from receptor-containing structures, which traffic into the perinuclear region (Oakley et al., 2000).

As both recycling (Gu et al., 2023; Mueller et al., 2002) and degradation (Gu et al., 2023) has been observed during CCR5 internalization, the cellular fate of the CCR5-arrestin2 complex was probed using the lysosomal-associated membrane protein 1 (LAMP1) as a marker of the lysosomal membranes (**Figure 4G**). No recruitment of the complex to the lysosomes (LAMP1, yellow) was detected in the presence of [6P4]CCL5 even 30 min after ligand stimulation. Similar results were obtained for both CCL5 and [6P4]CCL5 in live cells using the LysoTrackerTM dye that accumulates in acidic environments, such as the lysosome (**Figure 4-figure supplement 1B, C**). These results indicate that the CCR5 internalization pathway, although mediated by arrestin, does not lead to rapid receptor degradation in lysosomes.

Arrestin interaction with AP2 is essential for CCR5 internalization

The interactions of arrestin with clathrin and AP2 have mostly been studied for the internalization of the β 2-adrenergic (β 2AR) (Barsi-Rhyne et al., 2022; Beaudrait et al., 2017; Goodman et al., 1996; Kang et al., 2009; Kim and Benovic, 2002) and V2R (Barsi-Rhyne et al., 2022; Beaudrait et al., 2017) receptors. Here, we investigated how both interactions affect the internalization of CCR5 in a HeLa KO cell line (HeLa^{-arr2/3}) devoid of arrestin2 and arrestin3 as well as in HeLa (CCL2) cells (**Figure 5**, **Figure 5-figure supplement 1**, respectively).

To characterize the arrestin2-clathrin interaction, we generated a YFP-tagged arrestin2 construct with the clathrin-binding motif (arrestin2 ^{Δ LIELD}-YFP) removed. No significant difference in CCR5 internalization was detected in either HeLa cell line between wild-type arrestin2-YFP and arrestin2 ^{Δ LIELD}-YFP, when the cells were stimulated with [6P4]CCL5 (HeLa^{-arr2/3}: **Figure 5A-C**, HeLa: **Figure 5-figure supplement 1A**). Apparently, the arrestin2-clathrin interaction does not play a significant role in the internalization of the arr-class B CCR5 receptor. This is in contrast to findings for the arr-class A receptor β 2AR where removal of the LIELD motif impairs receptor internalization (Kim and Benovic, 2002).

To test the influence of the arrestin2-AP2 interaction, we introduced mutations within the arrestin2 adaptin-binding motif, which is conserved across all arrestins (**Figure 1-figure supplement 1A**). The mutation R395A within this motif has been found to inhibit β 2AR internalization by arrestin3 (Laporte et al., 2000). Introducing the equivalent mutation into a fluorescent arrestin2 construct (arrestin2^{R395A}-YFP), did not lead to significant changes in CCR5 internalization relative to wild-type arrestin2-YFP (HeLa^{-arr2/3}: **Figure 5A-C**, HeLa: **Figure 5-figure supplement 1B**).

The structure of AP2 β 2 in complex with the C-terminal arrestin2 peptide (Schmid et al., 2006) shows arrestin2 residues D385, F388, F391, and R395 interacting via salt bridges and base stacking with AP2 β 2 (**Figure 3A**). Apparently, these interactions lead to a rather stable complex of full-length arrestin2 with AP2 β 2⁷⁰¹⁻⁹³⁷ in solution as evidenced by our SEC binding assay, NMR and the protection of activated arrestin2 against proteolysis (**Figure 3**, **Figure 3-figure supplement 3**). We hypothesized that due to this extended arrestin2-AP2 β 2 interaction interface, the mutation of a single interacting residue may not suffice to inhibit the arrestin2-AP2 interaction leading to CCR5 internalization. Hence, we generated a further arrestin2 mutant (arrestin2 ^{Δ AP2}-YFP) with all four direct interacting residues replaced (D385A, F388A, F391A, R395A). Similar to

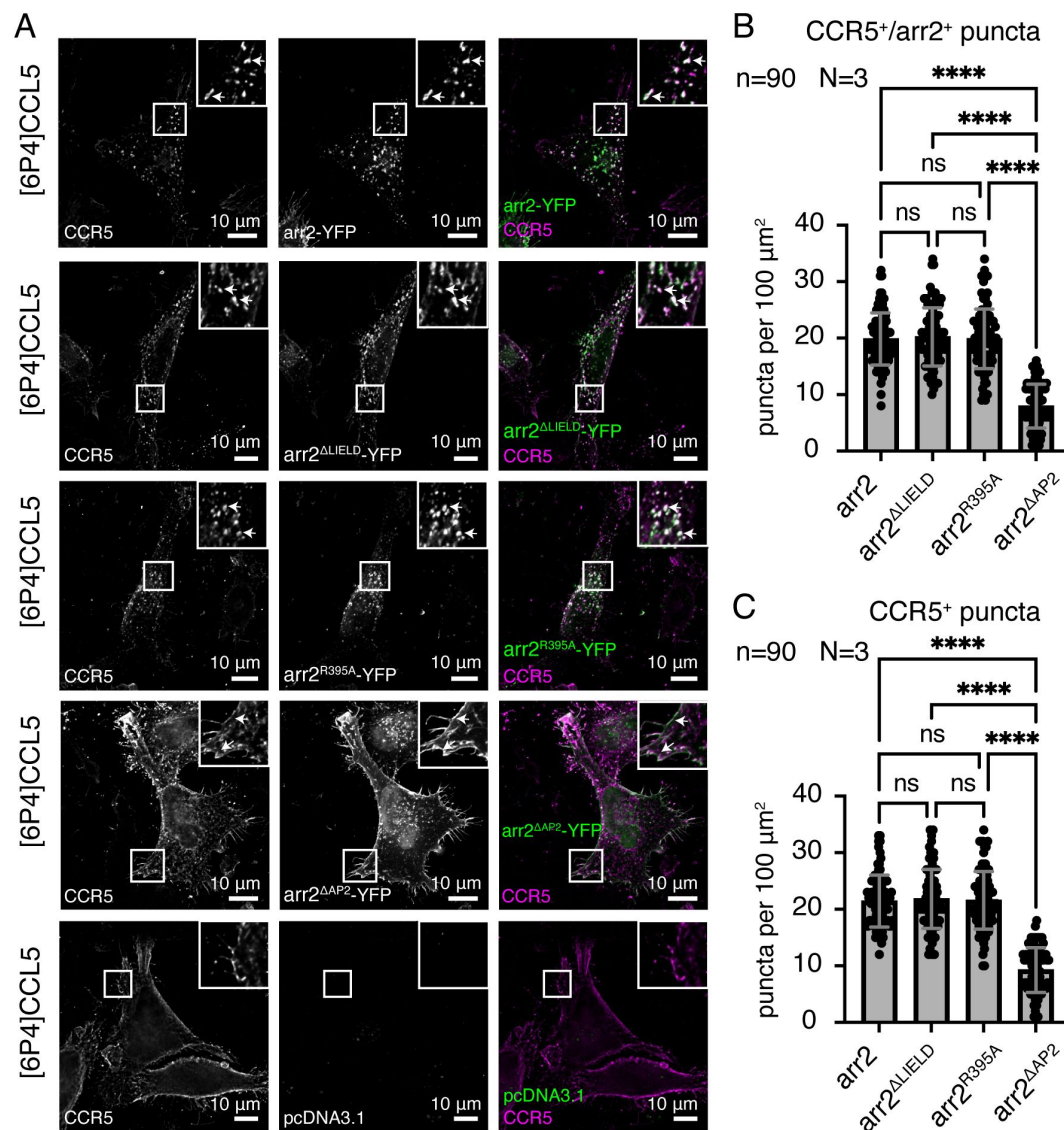


Figure 5

Dependence of CCR5 internalization on arrestin2 interactions with clathrin or AP2 monitored in HeLa^{-arr2/3} cells.

(A) CCR5 internalization induced by 60 min [6P4]CCL5 ligand stimulation in HeLa cells co-transfected with plasmids containing CCR5 and arrestin2-YFP, arrestin2-YFP^(LIED), arrestin2-YFP^{R395A}, arrestin2-YFP^(AP2) or empty pcDNA3.1. (B, C) Internalization was quantified by counting the number of puncta (B) positive for CCR5 and arrestin2 (CCR5⁺/arrestin2⁺) or (C) only CCR5⁺ 60 min after ligand stimulation in the HeLa^{-arr2/3} cells. No significant difference in CCR5 internalization is detected for the arrestin2-YFP^(LIED) and arrestin2-YFP^{R395A} constructs (ns, not significant: $P > 0.9999$) whereas the absence of the AP2 binding motif in the arrestin2^(AP2)-YFP construct causes a significant (****: $P < 0.0001$) reduction of CCR5 internalization 60 min after ligand incubation. Mean and standard deviation are shown for N=3 biological replicates and n=90 ROIs from 30 cells.

wild-type arrestin2-YFP, arrestin2^{ΔAP2}-YFP is recruited to CCR5 at the plasma membrane (HeLa: **Figure 5-figure supplement 1C** [↗](#)). However, the subsequent CCR5 internalization is potentially inhibited (HeLa^{-arr2/3}: **Figure 5A-C** [↗](#), HeLa: **Figure 5-figure supplement 1D** [↗](#)) as indicated by a reduction of the Mander's CCR5/arrestin2 colocalization coefficient (from ~0.8 to ~0.2) the number of puncta positive for CCR5 and arrestin2 (from ~20–13 to ~8–6 per 100 μm², similar values obtained for CCR5 puncta). In agreement with this finding, the arrestin2^{ΔAP2} mutant is unable to form a complex with AP2 in the presence of CCR5pp6 as assayed by SEC (**Figure 5-figure supplement 1E** [↗](#)). Taken together, these results show that the stable arrestin-AP2 interaction is crucial for CCR5 internalization, whereas the arrestin-clathrin interaction is not critical for sequestration from the plasma membrane.

Discussion

In the present study, we have elucidated the role of CCR5 phosphorylation in arrestin-clathrin and arrestin-AP2 interactions which are essential for CCR5 sequestration into cells. NMR and chromatographic data on arrestin2 interactions with clathrin and AP2 proteins in the presence of arrestin2-activating CCR5 C-terminal phosphopeptides revealed the molecular details underlying CCR5 sequestration. Whereas the analysis of arrestin2 ¹H-¹⁵N resonances was difficult due to significant line broadening from intrinsic microsecond dynamics (Isaikina et al., 2023 [↗](#)), the better quality of the clathrin-N ¹H-¹⁵N spectra allowed a detailed characterization of the arrestin2-clathrin interaction from the clathrin side. Arrestin2 binds to clathrin with about 100 μM affinity. This interaction is independent of arrestin2 activation by phosphopeptides, which may enable the internalization of receptors with low phosphorylation levels. The interaction is mediated through a single binding site in a 1:1 stoichiometry involving the arrestin2 clathrin-binding loop and the edges of blade 1 and blade 2 of clathrin-N. This binding site agrees with one of the binding sites observed in the crystal structure of the arrestin2-clathrin-N complex (PDB: 3GD1), but disagrees with the second crystal interaction site involving the arrestin2 splice loop. Presumably, the latter is a result of crystal packing. Our findings are supported by a previous study (Lally et al., 2017 [↗](#)), which suggests that the arrestin C-domain edge containing the splice loop anchors arrestin to the membrane. The membrane anchoring seems mutually exclusive with clathrin binding (Lally et al., 2017 [↗](#)).

In contrast to the arrestin2-clathrin interaction, the arrestin2-AP2 interaction requires arrestin2 activation by GPCR C-terminal tail phosphopeptides and depends quantitatively on their phosphorylation levels. The formed arrestin2-phosphopeptide-AP2 complexes are stable and more resistant to proteolysis than binary arrestin2-phosphopeptide complexes. The dissociation constant of the arrestin2-AP2 complex is about 10 μM in the presence of the fully phosphorylated CCR5pp6, which is a 10-fold higher affinity than that of the arrestin2-clathrin complex. This may indicate a preference towards AP2 complex formation for highly phosphorylated receptors. However, as the phosphorylation decreases the arrestin2-AP2 dissociation constant weakens to about 40 μM, making it more comparable to the arrestin2-clathrin interaction.

Cellular experiments demonstrated chemokine agonist-induced recruitment of arrestin2 towards CCR5 at the plasma membrane, which is followed by the sequestration of the CCR5-arrestin2 complex into the cytoplasm including the perinuclear region where it remains stable for an extended period. This provides experimental evidence that CCR5 belongs to the arr-class B of GPCRs as surmised previously from the large number of phosphorylation sites in its C-terminus and the presence of the pXpp motif (Isaikina et al., 2023 [↗](#)). The sequestered receptor remained stable inside the cells for at least 30 min, and was not rapidly degraded in lysosomes, even in the presence of [6P4]CCL5. A similar CCR5 lysosome-evading internalization pattern has been observed for the potent anti-HIV chemokine analog PSC-RANTES (Escola et al., 2010 [↗](#)). As a recent CCR5 trafficking study demonstrated that CCR5 scavenges both CCL5 and [6P4]CCL5 chemokines in an arrestin-dependent manner (Gu et al., 2023 [↗](#)), it is possible that the long retention of the CCR5-

arrestin complex and the ability of the receptor to undergo recycling (Gu et al., 2023 [↗](#); Mueller et al., 2002 [↗](#)) is utilized to remove the extracellular ligands from the cell surface. This may help in the regulation of chemokine ligand concentrations (Ariel et al., 2006 [↗](#)).

Based on the present CCR5 data, we envision the following CME mechanism, which may also apply to other arr-class B receptors (**Figure 6** [↗](#), left). The high receptor phosphorylation leads to a strong recruitment of arrestin even in the absence of interactions with negatively charged PIP₂ in the plasma membrane (Janetzko et al., 2022 [↗](#)). The consequent tight binding of the phosphorylated receptor tail to arrestin strand β 1 stably activates arrestin thereby releasing its strand β 20 and C-terminal tail. The freed β 20 strand can then interact with the β 2 subunit of AP2, which becomes released from the AP2 core upon AP2 activation by PIP₂ molecules (Kelly et al., 2014 [↗](#)). Apparently, the interaction of arrestin2 strand β 20 with the AP2 β 2 subunit is rather strong, since it could only be inhibited in cellular assays by a fourfold, but not by a single mutation in the arrestin2 C-terminal region. In contrast, removal of the clathrin-binding LIELD motif from arrestin2 did not impair receptor internalization, which is in agreement with the determined relatively low affinity of the arrestin2-clathrin interaction. This suggests that the clathrin interaction is less important for receptor internalization than the AP2 interaction.

The AP2 activation by PIP₂ also exposes the AP2 clathrin-binding LLNLD motif, responsible for clathrin recruitment and formation of the clathrin-coated pits (Kelly et al., 2014 [↗](#)). In the stable ternary complex of a highly phosphorylated arr-class B receptor with arrestin and AP2, the exposed AP2 LLNLD clathrin-binding motif may then subsequently couple to the clathrin network thereby inducing rapid sequestration from the plasma membrane into endosomes. The direct clathrin interaction of arrestin via its LIELD motif appears less important in this respect.

Although the generalization of this mechanism from CCR5 to other arr-class B receptors has to be explored further, it is indirectly corroborated in the visual rhodopsin-arrestin1 system. The arr-class B receptor rhodopsin (Isaikina et al., 2023 [↗](#)) also undergoes CME (Moaven et al., 2013 [↗](#)) with arrestin1 harboring the conserved AP2 binding motif, but missing the clathrin-binding motif (**Figure 1-figure supplement 1A** [↗](#)). As arr-class B receptors are almost all peptide-binding GPCRs (Isaikina et al., 2023 [↗](#); Oakley et al., 2000 [↗](#)), we speculate that the strong arrestin interaction may also be utilized to remove the extracellular ligands from the plasma membrane, independent of receptor degradation.

In comparison to arr-class B receptors, arr-class A receptors have a lower phosphosite density in their C-terminal tails. Previous studies on the arr-class A β 2AR (Kang et al., 2009 [↗](#); Kim and Benovic, 2002 [↗](#); Laporte et al., 2000 [↗](#)) revealed that both clathrin and AP2 interaction are crucial for internalization. It is expected that the lower phosphosite density of arr-class A receptors results in a less stable interaction with arrestin (**Figure 6** [↗](#), right). Therefore additional interactions with PIP₂ lipids are required to anchor arrestin to the membrane and play a significant role in receptor endocytosis (Janetzko et al., 2022 [↗](#)). As a consequence of the low phosphorylation of the receptor and the weakened interaction with arrestin strand β 1, the equilibrium between inactive (strand β 20 forms β -sheet with β 1) and active (β 20 released) arrestin is shifted towards the inactive conformation, even when arrestin is bound to the receptor via core contacts. Due to the low availability of strand β 20, the arrestin interaction with AP2 is considerably weakened and comparable in strength to the clathrin interaction. This is consistent with the finding that AP2 and clathrin participate with similar strength in arr-class A receptor internalization (Kang et al., 2009 [↗](#); Kim and Benovic, 2002 [↗](#); Laporte et al., 2000 [↗](#)). However, as both interactions are weak, the sequestration is reduced. In addition, the weaker receptor phosphorylation leads to fast dissociation of arrestin from the receptor at or near the plasma membrane, and in consequence the receptor traffics without arrestin into endosomes (Oakley et al., 2001 [↗](#), 2000 [↗](#)).

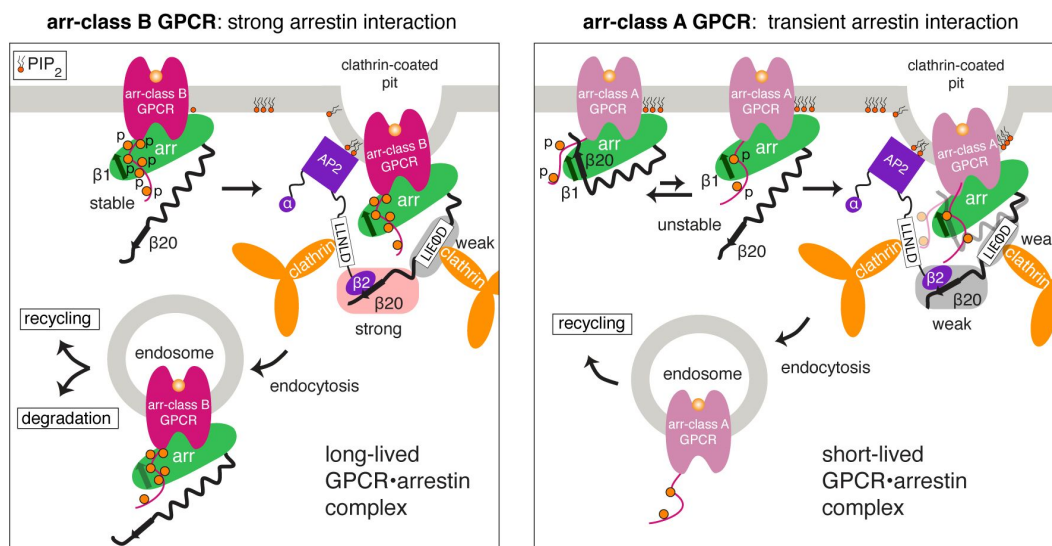


Figure 6

Overall scheme of arr-class A and B GPCR internalization.

Schematic difference of arrestin2-mediated internalization of arr-class B (left) vs. arr-class A (right) GPCRs. Arr-class B GPCRs bind stably to arrestin due to their high levels of phosphorylation. This results in a robust release of the arrestin C-terminus, a stable interaction with AP2 and formation of a long-lived GPCR-arrestin complex. Arr-class A GPCRs bind weakly to arrestin due to their poor phosphorylation. They require stabilization of the arrestin complex by membrane-bound PIP₂ molecules. The arrestin C-terminus is not fully released and consequently the interaction with AP2 is unstable.

In conclusion, this biophysical and cellular analysis of the interactions between the CCR5, arrestin2, and clathrin/AP2 endocytic machinery elucidates the interplay of molecular forces driving GPCR internalization. The strength of the receptor phosphorylation modulates the interaction with arrestin2, the release of its strand β 20, and subsequently its interaction with the β 2 subunit of AP2. In comparison, the arrestin2-clathrin interaction is weaker. The interaction between the receptor phosphosites and arrestin appears as a distinctive factor between arr-class B and arr-class A receptor sequestration, since phosphorylation of the latter is considerably weaker. These findings expand on the previously suggested mechanism for the reduced sequestration of the arr-class A receptor β 2AR by Kim and Benovic (Kim and Benovic, 2002 [DOI](#)) and Laporte et al (Laporte et al., 2000 [DOI](#)), where the weakened AP2-arrestin and the phosphorylation-independent clathrin-arrestin interactions appear of similar importance. The phosphorylation dependence of the AP2-arrestin interaction may also play a role in modulating the strength of downstream arrestin-mediated signaling from endosomes (Jean-Charles et al., 2017 [DOI](#); Tohgo et al., 2003 [DOI](#)). A detailed understanding of these molecular interactions may provide a framework for the development of novel therapeutics interfering with GPCR sequestration.

Materials and Methods

Constructs

The genes encoding full-length (arrestin2¹⁻⁴¹⁸) and truncated (arrestin2¹⁻³⁹³) human arrestin2 constructs have been described before (Isaikina et al., 2023 [DOI](#)). The arrestin2^{ΔCBL} construct in a pET-28a (+) vector was obtained from the GenScript by replacing the loop residues of arrestin2 K357 to D385 with visual arrestin residues E365 to N376. The construct also contained an N-terminal hexahistidine tag for purification followed by a TEV cleavage site (ENLYFQG), resulting in the following total sequence (arrestin2^{ΔCBL} in bold, visual arrestin loop underlined):

MGSSHHHHHHSSGENLYFQGMGDKGTRVFKKASPNGKLTVYLGKRDFVDHIDLVPDVGVLVDPEYLKERRVYVT
LTCAFRYGREDLDVLGLTFRKDLFVANVQSFPAPEDKKPLTRLQERLIKKLGEHAYPFTFEIPPNLPCSVTLQPG
PEDTGKACGVVDYEVKAFLAENLEEKIKHRNSVRLVIRKVQYAPERPGPQPTAETTRQFLMSDKPLHLEASLDKEIY
YHGEPISVNVHVTNNTNKTVMKKIKISVRQYADIVLFNTAQYKVPVAMEEADDTVAPSSSTFSKVYTLTPFLANNREK
RGLALDGKLKHEDTNLASSTLLREGANREILGIIVSYKVVKLVVSRGGLGDLASSDVAVELPFTLMHPKPEDPA
KESYQDANIVFEDFARQLKGMKDDKEEEDGTGSPQLNNR

clathrin-N [pETM33_CLTC_Clathrin propeller, clathrin heavy chain 1 residues 1-364, Addgene plasmid #178471 (Benz et al., 2022 [DOI](#))] and AP2 β ⁷⁰¹⁻⁹³⁷ [pETM33_AP2B1_adaptin, Addgene plasmid #178467 (Benz et al., 2022 [DOI](#))] constructs were gifts from Ylva Ivarsson. Arrestin2-YFP was a gift from Robert Lefkowitz [Addgene plasmid # 36916, (Violin et al., 2006 [DOI](#))].

The arrestin2^{ΔLIED}-YFP construct was obtained from the arrestin2-YFP template by standard PCR reaction using KAPA HiFi polymerase (forward primer: 5'-GAG ACT CCA GTA GAC ACC AAT ACC AAT GAT GAC GAC A-3'; reverse primer: 5'-TGT CGT CAT CAT TGG TAT TGG TGT CTA CTG GAG TCT C-3').

Arrestin2^{R395A}-YFP was generated in the same manner using forward primer: 5'-GGA CTT TGC TCG TCA GGC GCT GAA AGG CAT GAA G-3' and reverse primer: 5'-CTT CAT GCC TTT CAG CGC CTG ACG AGC AAA GTC C-3'.

The same protocol was applied for generating the arrestin2^{ΔAP2}-YFP clone with the forward primer 5'-GAC ACC AAT GAT GAC GCC ATT GTG GCT GAG GAC GCT GCT CGT CAGGCG CTG AAA GGC ATG-3' and reverse primer 5'-CAT GCC TTT CAG CGC CTG ACG AGC AGC GTC CTC AGC CAC AAT GGC GTC ATC ATT GGT GTC-3'.

The CCR5 construct in pcDNA3.1 used in cellular assays was obtained from GenScript. The complete construct contained the full-length human CCR5 sequence (bold) followed by a GS- linker and a C-terminal FLAG-tag (underlined):

MDYQVSSPIYDINYTTSEPCQKINVKQIAARLLPPLYSLVFIFGFVGNMLVILILINCKRLKSMTDIYLLNLAISD
LFFLLTVPFWAHYAAAQWDFGNTMCQLLTGLYFIGFFSGIFFIILLTIDRYLAVVHAVFALKARTVTFGVVTSVIT
WVAVFASLPGIIFTRSQKEGLHYTCSSHPYSQYQFWKNFQTLKIVILGLVPLLVVICYSYGILKTLLRCRNEK
KRHRVRLIFTIMIVYFLFWAPYNIVLLLNTFQEFGNLCSSSNRLDQAMQVTETLGMTHCCINPIIYAFVGEKF
RNYLLVFFQKHIARFCKCCSIFQQEAPERASSVYTRSTGEQEISVGLSGGGGGSGSSSGGGSGSSSGGEFDY
KDDDDK

The genes encoding [5P12]CCL5 (Wiktor et al., 2013 [DOI](#)), CCL5 (Wiktor et al., 2013 [DOI](#)) and [6P4]CCL5 (Isaikina et al., 2022 [DOI](#)) for expression in *E. coli* have been described previously.

Protein expression and purification

All protein constructs were expressed in *E. coli* BL21 (DE3) cultured in Lysogeny broth medium (LB). For the preparation of the deuterated ^{15}N -labeled (and ^{13}C -labeled) NMR samples, cells were grown in $\text{D}_2\text{O}/^{15}\text{NH}_4\text{Cl}$ (^{13}C -glucose) M9 minimal medium. For the preparation of Ile- δ - $^{13}\text{CH}_3$ -labeled arrestin2¹⁻³⁹³, the minimal medium was supplemented with 80 mg per liter of 2-ketobutyric acid-4- ^{13}C , 3,3- d_2 sodium salt hydrate following previously reported procedures (Tugarinov et al., 2006 [DOI](#)). In all cases, cells were grown at 37°C until the optical density at 600 nm ($\text{OD}_{600\text{ nm}}$) reached 0.7–0.8. Thereafter protein expression was induced by the addition of 25 μM isopropyl β -D-thiogalactopyranoside (IPTG) (arrestin2 constructs) or 1 mM IPTG (clathrin-N and AP2 β ⁷⁰¹⁻⁹³⁷). The temperature was lowered to 18°C (arrestin2 constructs) or 22°C (clathrin-N and AP2 β ⁷⁰¹⁻⁹³⁷) for an overnight expression, after which the cells were harvested by centrifugation (30 min at 6000 rpm).

Arrestin2

The purification of arrestin2 has been described before (Isaikina et al., 2023 [DOI](#)). In short, arrestin2¹⁻⁴¹⁸, arrestin2¹⁻³⁹³, arrestin2 ^{ΔCBL} and arrestin2 ^{$\Delta\text{AP}2$} were purified on a Ni-NTA HiTrap HP column (GE Life Sciences) and the His-tag was removed by overnight cleavage with TEV protease (homemade). The cleaved protein was further separated from impurities by a reverse IMAC step on a Ni-NTA HiTrap HP column. An ion exchange chromatography step using a 5-ml HiTrap Q HP anion exchange column was then carried out before the final gel filtration step on a HiLoad 16/600 Superdex 200 pg gel filtration column (GE Healthcare) equilibrated with 20 mM HEPES, 150 mM NaCl, pH 7.4 (SEC buffer I). Protein purity was confirmed by SDS-PAGE.

clathrin-N

clathrin-N was purified similarly to the arrestin2 protocol using a Ni-NTA HiTrap HP column. The His-tag was removed by overnight cleavage with 3C protease (homemade), followed by a reverse IMAC step. The protein was concentrated in a Vivaspin 20 (Sartorius) concentrator (10-kDa MWCO) before a final gel filtration step on a HiLoad 16/600 Superdex S75 pg gel filtration column (GE Healthcare) equilibrated with SEC buffer I. Protein purity was confirmed by SDS-PAGE.

AP2 β 2

AP2 β 2 was purified similarly to clathrin-N. Due to a change of the isoelectric point from 6.6 to 7.7 after 3C protease cleavage, the pH of the SEC buffer was changed to 7.0 (SEC buffer II, 20 mM HEPES, 150 mM NaCl, pH 7.0).

Determination of protein melting temperature

Protein melting temperatures were determined routinely by differential scanning fluorimetry using a Prometheus Nanotemper NT.48 (Tecan) instrument.

Peptide synthesis

Phosphorylated peptides (**Table S1** [↗](#)) corresponding to the last 22 residues of the human CCR5 receptor (CCR5pp3, CCR5pp4, CCR5pp6) were obtained from the Tufts University Core Facility for peptide synthesis.

Cell culture

HeLa (ATCC CCL2) cells and HeLa KO cell line (HeLa^{-arr2/3}) ([D'Agostino et al., 2020](#) [↗](#)) devoid of arrestin2 and arrestin3 were grown in high-glucose Dulbecco's modified Eagle's medium (Sigma-Aldrich) with 10% fetal bovine serum (FBS, Biowest), 2 mM L-glutamine (#25030; Gibco), 100 U ml⁻¹ penicillin G and 100 ng ml⁻¹ streptomycin (P433; Sigma-Aldrich Chemie GmbH), 1 mM sodium pyruvate (#S8636; Sigma-Aldrich Chemie GmbH) at 37°C and 5% CO₂. For transient cell transfections, cells were plated into six-well plates to reach 70% confluency the following day and transfected with 0.3 µg of each arrestin2-YFP and CCR5 plasmid DNA complexed with Helix-IN transfection reagent (OZ Biosciences).

Immunostaining and microscopy

In HeLa cell lines, arrestin2 and CCR5 were overexpressed by transiently transfecting arrestin2-YFP and CCR5 respectively. CCR5 was stained with a primary FLAG antibody (1:200, Thermo Fisher F3165) in 5% FBS in PBS. A secondary mouse (1:500) Alexa 633 (A21052, Invitrogen) antibody was used in 5% FBS in PBS. Fluoromount-G mounting medium (Thermo Fischer) was used to mount the coverslips onto the imaging slide. Fluorescence and DIC images were acquired with an ORCA-Flash 4.0 camera (Hamamatsu) mounted on an Axio Imager.M2 fluorescence microscope with a 63x Plan-Apochromat objective (Carl Zeiss) and an HXP 120C light source with ZEN 2.6 software. Image processing was performed using OMERO.insight client ([Allan et al., 2012](#) [↗](#)), and analyzed with Fiji software ([Schindelin et al., 2012](#) [↗](#)).

To assess the recruitment of arrestin2 by CCR5, arrestin2-YFP- and CCR5-expressing cells were incubated with 1 µM chemokine ligands. Individual coverslips were removed and fixed with 4% PFA for 10 min at 0, 10, and 30 min post-ligand stimulation. Fixed cells were permeabilized with 0.1% Triton X-100 in PBS for 5 min, rinsed with PBS twice and blocked with 5% FBS in PBS. CCR5 was stained as mentioned previously and imaged. Colocalization between arrestin2 and CCR5 was assessed by measuring the Mander's coefficient in 2-3 regions of interest (ROI) of 100 µm² per cell using the Fiji JACOP plug-in ([Bolte and Cordelières, 2006](#) [↗](#)). The number of internalized CCR5 puncta with and without arrestin2 was quantified manually by taking 3 ROIs of 100 µm² per cell using Fiji.

To image arrestin2/CCR5 trafficking to lysosome, arrestin2-YFP and CCR5 expressing cells were treated with [6P4]CCL5. Individual coverslips were removed and fixed with 4% PFA for 10 min at 0, 10, and 30 min post stimulation. Fixed cells were permeabilized with 0.1% Triton X-100 in PBS for 5 min, rinsed with PBS twice and blocked with 5% FBS in PBS. Lysosomes were stained with LAMP1 antibody (1:200, Cell Signaling Technology, D2D11). CCR5 was stained with FLAG antibody (1:200, Thermo Fisher F3165). Secondary rabbit (1:500) Alexa-Fluor 568 (A110042 Invitrogen) and secondary mouse (1:500) Alexa-Fluor 633 (A21052, Invitrogen) fluorescent antibodies were used respectively. Fixation, imaging and analysis were performed as described above.

For assessing the effect of arrestin2 mutations, respective mutants were overexpressed with CCR5 in HeLa^{-arr2/3} and HeLa cells, followed by immunostaining and microscopy analysis as described above (HeLa^{-arr2/3}: 60 min, HeLa: 30 min post [6P4]CCL5 stimulation).

Live-cell imaging

For live-cell imaging, HeLa cells expressing arrestin2-YFP were seeded on an ibidi μ -slide VI 0.4 channel slide (#80606, ibidi), 24 hours prior to data acquisition. Live cell imaging was performed in complete growth medium lacking phenol red at 37°C with 5% CO₂ 508 using an inverted Axio Observer microscope (Zeiss) with a Plan Apochromat N 63x/1.40 oil DIC 509 M27 objective and a Photometrics Prime 95B camera. Filters with standard specifications for GFP were used to image arrestin2-YFP. After selecting cells to be imaged, images were acquired at 5 min intervals. Image processing was performed using OMERO.insight client (Allan et al., 2012 [↗](#)), and analyzed with Fiji software (Schindelin et al., 2012 [↗](#)).

To image lysosome stained with lysotracker, HeLa cells expressing arrestin2-YFP and CCR5 were seeded on an ibidi μ -slide VI 0.4 channel slide (#80606, ibidi), 24 hours prior to data acquisition. LysoTracker™ Deep Red (L12492) and peptide ligand was added in complete growth medium lacking phenol red. Images were taken at 0 min and 30 min.

Statistics

Statistical analysis was performed using GraphPad Prism 10.1.1 (GraphPad Software, Inc., San Diego, CA, USA). For statistical evaluation, the normality of the data was routinely tested using the Shapiro-Wilk or Kolmogorov-Smirnov normality test. For the comparison of two groups (**Figure 5A** [↗](#), B), a t-test was employed, whereas to compare three or more groups (**Figure 4D** [↗](#)), a one-way ANOVA was performed followed by a Kruskal-Wallis test.

NMR Experiments

NMR samples were prepared in SEC buffer I supplemented with 5% D₂O, 2 mM EDTA and 0.03% NaN₃ as 270- μ l volumes in Shigemi tubes. The assignment sample contained 300 mM NaCl. The spectra were recorded on a Bruker AVANCE 14.1 T (600 MHz) spectrometer (arrestin2¹⁻³⁹³, AP2 β ⁷⁰¹⁻⁹³⁷) or a Bruker AVANCE 21.2 T (900 MHz) spectrometer (clathrin-N) equipped with a TCI cryoprobe at 303 K. The spectra were processed with NMRPipe (Delaglio et al., 1995 [↗](#)), assigned using CcpNmr AnalysisAssign (Skinner et al., 2016 [↗](#)), and analyzed with NMRFAM-SPARKY (Lee et al., 2015 [↗](#)).

Arrestin2 assignment

For the assignment of arrestin2 backbone resonances, HNCACB-TROSY, HNCA-TROSY and HNCO-TROSY spectra were recorded on ¹⁵N-, ¹³C-, ²H-labeled arrestin2¹⁻³⁹³ (100-150 μ M) samples.

Arrestin2 titrations

For detecting the clathrin-N-arrestin2 interaction on arrestin2 by NMR, a titration of 50 μ M ¹⁵N-, ²H-labeled arrestin2¹⁻³⁹³ with unlabeled clathrin-N up to an equimolar amount was monitored by ¹H-¹⁵N HSQC-TROSY spectra recorded as 100 (¹⁵N) x 512 (¹H) complex points and acquisition times of 50 ms (¹⁵N) and 55 ms (¹H) for 4.5 h. Peak intensities *I* were determined by integration and the interaction was quantified relative to the intensity *I*₀ in the absence of clathrin-N. An intensity attenuation larger than one standard deviation of *I*/*I*₀ calculated from all resonances was considered significant.

Similarly, a titration of 50 μ M Ile- δ 1-¹³CH₃, ²H-labeled arrestin2¹⁻³⁹³ with unlabeled clathrin-N up to a concentration of 600 μ M was monitored by ¹H-¹³C HMQC spectra recorded as 32 (¹³C) x 512 (¹H) complex points and acquisition times of 20 ms (¹³C) and 55 ms (¹H) for 1 h. The assignments

of the methyl groups were transferred from BMRB ID:51131 (Shiraishi et al., 2021 [DOI](#)). K_D values were obtained by nonlinear least-squares fitting using Matlab (Matlab_R2021b, MathWorks, Inc.) and the following equation:

$$\Delta\delta = \Delta\delta_{\max} \frac{[L] + [P] + K_D - \sqrt{([L] + [P] + K_D)^2 - 4[P][L]}}{2[P]} \quad (1)$$

where $\Delta\delta = \delta_{\text{apo}} - \delta_{\text{bound}}$ is the difference in the ^{13}C chemical shift, $\delta_{\max}$ is the difference between apo and the fully ligand-bound state, and [P] and [L] are the total protein (arrestin2) and ligand (clathrin-N) concentrations, respectively.

Combined $^1\text{H}/^{13}\text{C}$ chemical shift perturbations $\Delta\delta_{\text{HC}}$ were calculated from the relation:

$$\Delta\delta_{\text{HC}} = \sqrt{\Delta\delta_{\text{H}}^2 + \left(\frac{\Delta\delta_{\text{C}}}{3.6}\right)^2} \quad (2)$$

where $\Delta\delta_{\text{H}}$ is the methyl proton and $\Delta\delta_{\text{C}}$ is the ^{13}C -methyl chemical shift difference between the apo and bound state. $\Delta\delta_{\text{HC}}$ values greater than one standard deviation from the mean were considered significant.

clathrin-N titrations

For detecting the clathrin-N-arrestin2 interaction on clathrin-N by NMR, a titration of 50 μM ^{15}N -, ^2H -labeled clathrin-N with unlabeled arrestin2¹⁻³⁹³, arrestin2¹⁻⁴¹⁸, arrestin2 ^{ΔCBL} or arrestin2¹⁻³⁹³•CCR5pp6 up to an equimolar amount was monitored by ^1H - ^{15}N HSQC-TROSY spectra recorded as 100 (^{15}N) x 1024 (^1H) complex points and acquisition times of 35 ms (^{15}N) and 80 ms (^1H) for 4.5 h. Assignments for clathrin-N were transferred from BMRB ID:25403 (Zhuo et al., 2015 [DOI](#)). Peak intensities were determined by quantifying peak volumes. Binding regions were determined by intensity ratios as described above.

AP2 β 2 titrations

For detecting the AP2 β 2-arrestin2 interaction on AP2 β 2⁷⁰¹⁻⁹³⁷ by NMR, a titration of 50 μM ^{15}N -, ^2H -labeled AP2 β 2⁷⁰¹⁻⁹³⁷ with unlabeled arrestin2¹⁻⁴¹⁸ or arrestin2¹⁻⁴¹⁸•CCR5pp6 up to an equimolar amount was monitored by ^1H - ^{15}N HSQC-TROSY spectra recorded as 100 (^{15}N) x 512 (^1H) complex points and acquisition times of 50 ms (^{15}N) and 55 ms (^1H) for 1 h. Assignments for AP2 β 2⁷⁰¹⁻⁹³⁷ were transferred from BMRB ID: 52320 (Naudi-Fabra et al., 2024 [DOI](#)). Peak intensities were determined by quantifying peak volumes. Binding regions were determined by intensity ratios as described above.

SEC binding assay

SEC binding assays were performed on an UltiMate 3000 HPLC system. Samples containing single polypeptides or their mixtures as indicated in the text and figure legends (arrestin2¹⁻⁴¹⁸: 10 μM , AP2 β 2: 10 μM , clathrin-N: 10 μM , CCR5pp0: 100 μM , CCR5pp3: 100 μM , CCR5pp4: 100 μM , CCR5pp6: 25-100 μM) were prepared as 270 μL volumes in SEC buffer II. Samples were injected with an autosampler using a 250- μL sample-loop onto a self-packed 4.2-ml S200 10/300 SEC column (length 25 mm, diameter 4.6 mm), which had been pre-equilibrated with SEC buffer II. Complex formation was monitored by following protein absorbance at 280 nm at 2.6 ml elution volume and the composition of the respective SEC fractions of all prepared complexes was analyzed by SDS-PAGE (Figure 3-figure supplement 2C [DOI](#)).

The apparent dissociation constant of the arrestin2¹⁻⁴¹⁸-AP2 β 2⁷⁰¹⁻⁹³⁷ complex in the presence of the CCR5 phosphopeptides could be determined by integrating the SEC absorption peaks of the complex and of the apo proteins and using their respective extinction coefficients. The increase of

absorbance around 2.6 ml is a consequence of the additional AP2 β 2⁷⁰¹⁻⁹³⁷ in that elution region. This region overlaps to some extent with the elution profile of apo arrestin2¹⁻⁴¹⁸. Taking into account the 1:1 stoichiometry, the amount of formed complex was estimated by integrating the profile between 2.4 and 2.8 ml, subtracting the respective integral of the apo arrestin2¹⁻⁴¹⁸ profile, and dividing the difference by the AP2 β 2⁷⁰¹⁻⁹³⁷ extinction coefficient. Since the phosphopeptides do not contain tryptophan, their absorbance contribution was ignored.

Trypsin proteolysis

Trypsin proteolysis assays were carried out by pre-incubating 100 μ L of 20 μ M arrestin2¹⁻⁴¹⁸ and 100 μ M CCR5pp6 or 500 μ M CCR5pp4/CCR5pp3 phosphopeptide with 1- (CCR5pp6), 2- (CCR5pp6) and 5-molar equivalents (CCR5pp6, CCR5pp4, CCR5pp3) of AP2 β 2 protein at room temperature for 5-10 min in the SEC buffer II. Following pre-incubation, 1 ng of Trypsin Gold (Promega) was added, and the mixture further incubated at 35 °C under gentle shaking (500 rpm, ThermoMixer). Samples for SDS-PAGE (10 μ L mixed with 10 μ L of 4x SDS loading buffer) were taken at 0, 5, 10, 20, 30, 60, and 90 min. The reaction was quenched by boiling the samples for 10 min at 95°C. Samples were then loaded on 4-20% precast gradient gels and visualized with Instant Blue protein stain (Abcam). Control reactions were run with apo arrestin2¹⁻⁴¹⁸ and arrestin2¹⁻⁴¹⁸ incubated with 100 μ M CCR5pp6 or 60-500 μ M CCR5pp4/CCR5pp3 phosphopeptide.

SDS-PAGE gels were analyzed using ImageJ software (Schneider et al., 2012 [↗](#)) by integrating the peak area of the two bands corresponding to full-length arrestin2¹⁻⁴¹⁸ (top) and arrestin2¹⁻⁴¹⁸ cleaved at residue R393 (bottom). The relative abundance (in %) of arrestin2¹⁻⁴¹⁸ was calculated from the respective integrals at 5 min upon trypsin addition.

Supplementary Information

Tables

Phosphopeptide	Sequence	K _D (arrestin2 ¹⁻³⁹³) [μ M] ^a
CCR5pp6	APERASSVY <u>TR</u> STGEQEI <u>SV</u> GL	45 ± 6
CCR5pp4	APERASSVY <u>TR</u> STGEQEI <u>SV</u> GL	199 ± 10
CCR5pp3	APERASSVY <u>TR</u> STGEQEI <u>SV</u> GL	198 ± 6

^adata from Isaikina, Petrovic et al. (2023).

Table S1.

CCR5 phosphopeptide sequences and their affinities towards arrestin2.

Underlined serine or threonine residues are phosphorylated.

Figures

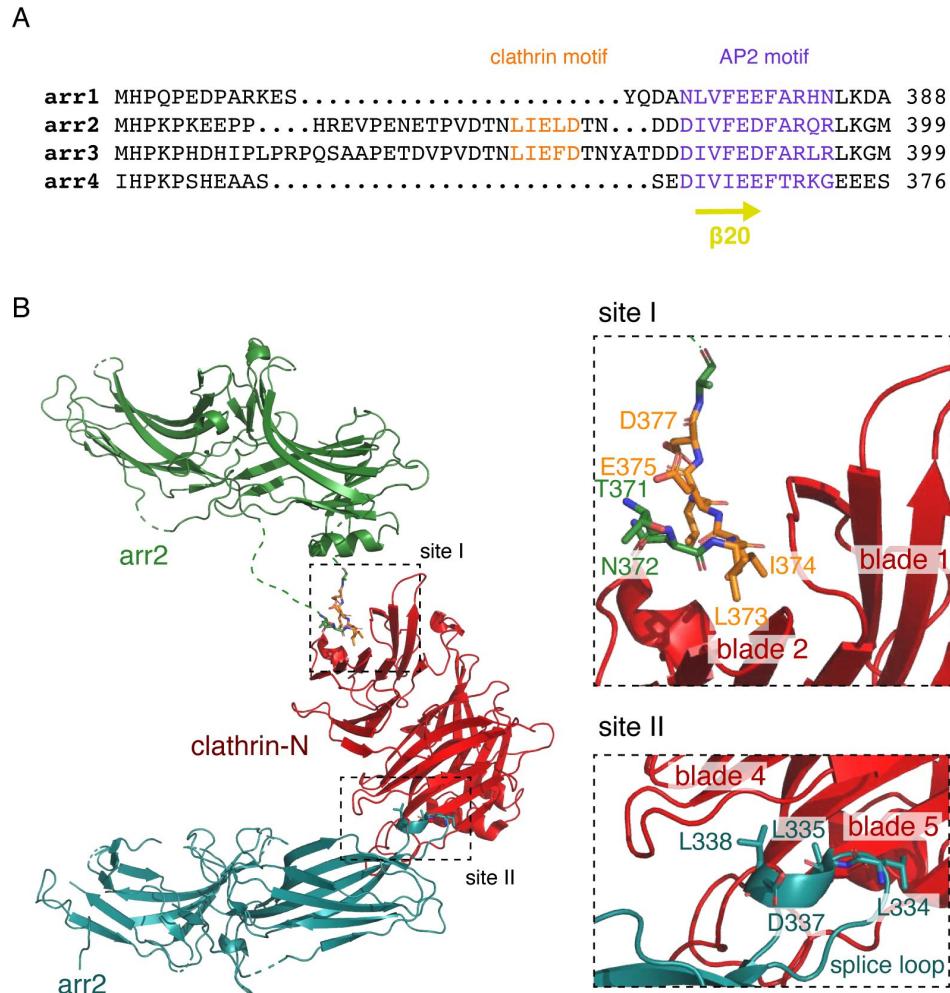


Figure 1-figure supplement 1.

Arrestin sequence alignment and analysis of the arrestin2-clathrin-N complex structure.

(A) Sequence alignment of four human arrestin types. The clathrin-binding motif (orange) and AP2 binding motif (purple) are indicated within the sequences. The numbering on the right corresponds to the last amino acid in the alignment. (B) X-ray structure of the arrestin2-clathrin complex (PDB:3GD1⁴⁵). The structure shows two arrestin chains (green and blue) interacting with one clathrin-N molecule (red). Interaction sites 1 and 2 are indicated by dashed rectangles with enlarged views on the right. The arrestin2 residues involved in the interaction are shown as sticks. Site I: between CBL of arrestin2 (green) and the edge of β -sheet blade 1 and 2 of clathrin-N. CBL residues are shown as sticks. The numbering follows the PDB deposition. Site II: between arrestin2 splice loop (blue) and β -sheet blade 4 and 5 of clathrin-N.

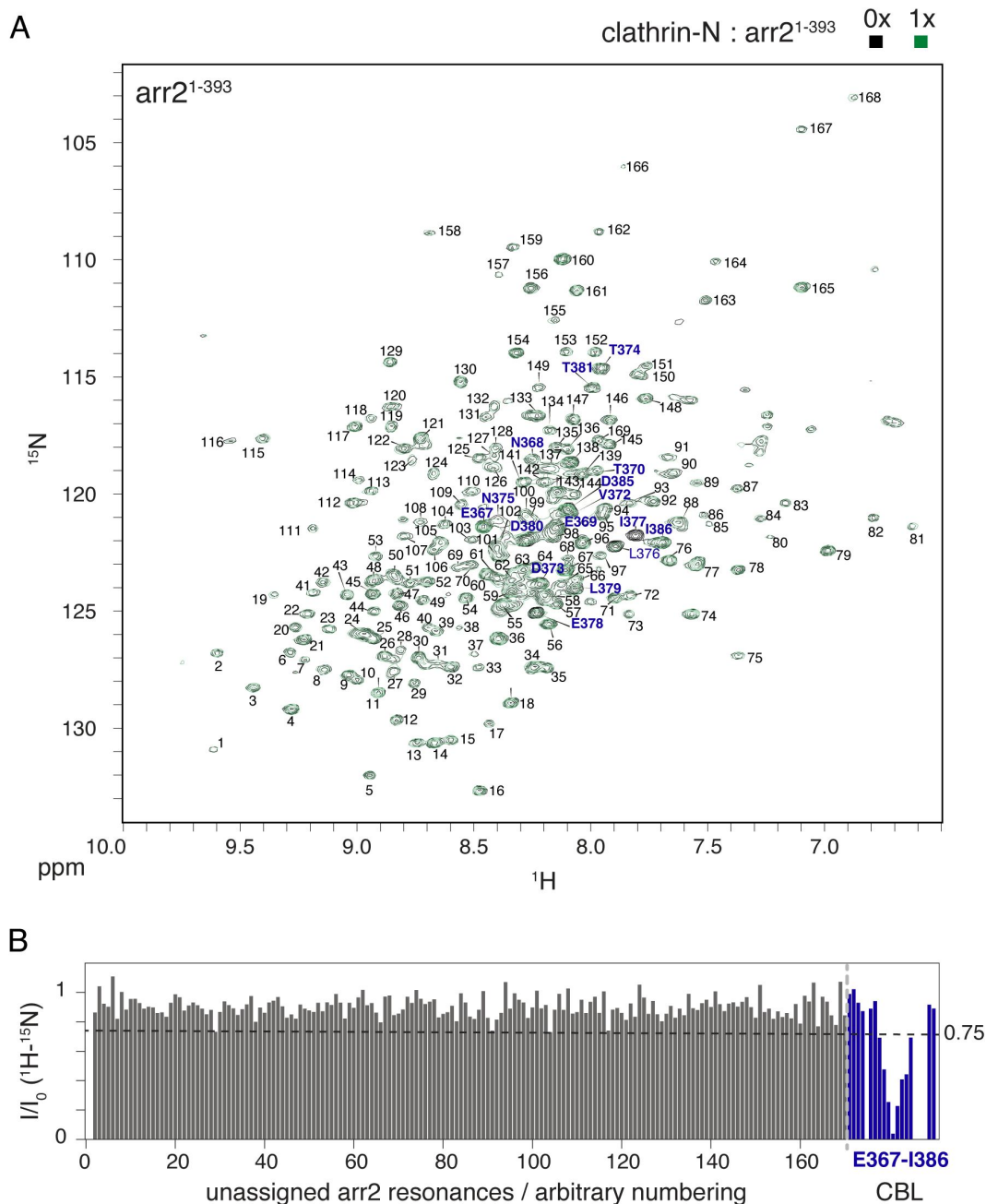


Figure 1-figure supplement 2.

Mapping of clathrin-N interaction on arrestin2.

(A) $^1\text{H}-^{15}\text{N}$ TROSY spectrum of arrestin2¹⁻³⁹³ in apo (black) or upon equimolar addition of clathrin-N (green) recorded on a Bruker AVANCE 14.1 T (600 MHz) spectrometer equipped with a TCI cryoprobe at 303 K. Assigned resonances are marked by blue amino acid numbers. Unassigned resonances are marked by black arbitrary numbers. (B) Intensity reduction I/I^0 of arrestin2¹⁻³⁹³ $^1\text{H}-^{15}\text{N}$ TROSY resonances upon clathrin-N interaction. The dashed line is drawn at one standard deviation below the average I/I^0 value. The resonance numbering follows panel (A).

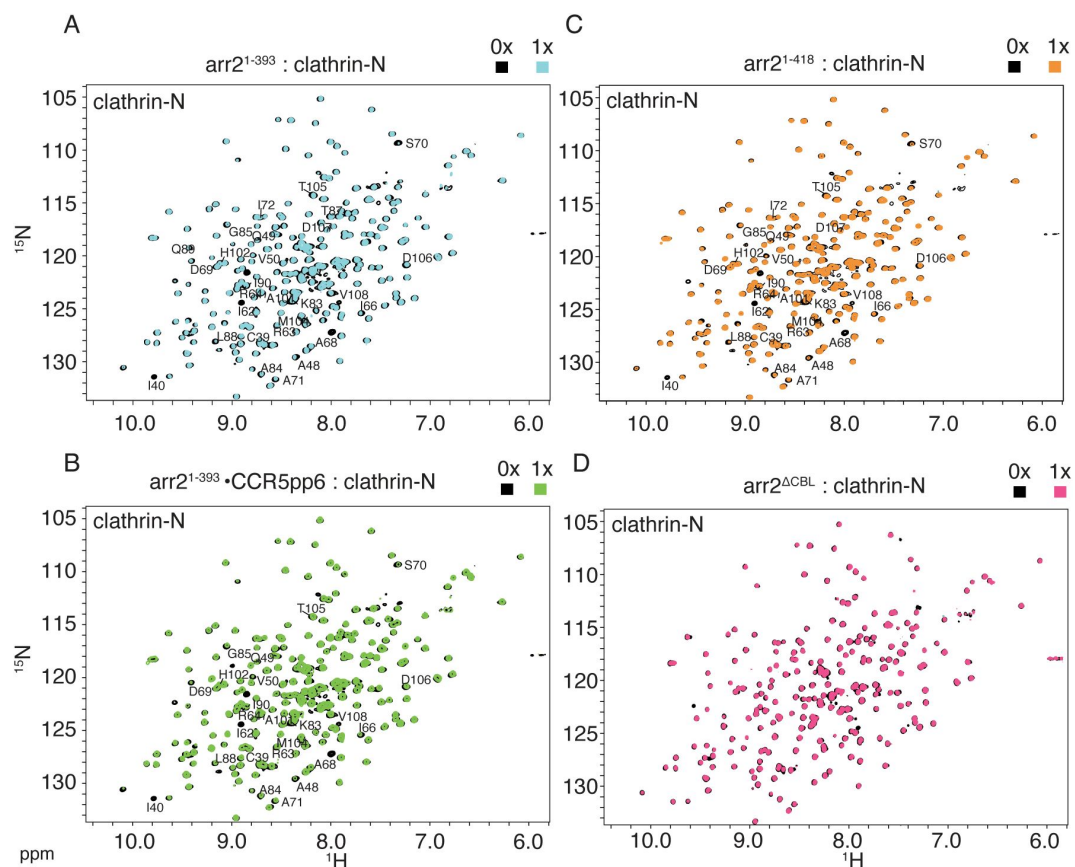


Figure 2-figure supplement 1.

Mapping of arrestin2 interaction on clathrin-N.

^1H - ^{15}N TROSY spectra of clathrin-N in the presence of one molar equivalent of (A) arrestin2 $^{1-393}$ (cyan); (B) arrestin2 $^{1-393}$ -CCR5pp6; (C) arrestin2 $^{1-418}$ or (D), arrestin2 $^{\Delta\text{CBL}}$ superimposed on ^1H - ^{15}N TROSY spectrum of apo clathrin-N (black). The spectra were recorded on a Bruker AVANCE 21.1 T (900 MHz) spectrometer equipped with a TCI cryoprobe at 303 K. Assigned residues undergoing significant line broadening are indicated on the spectra.

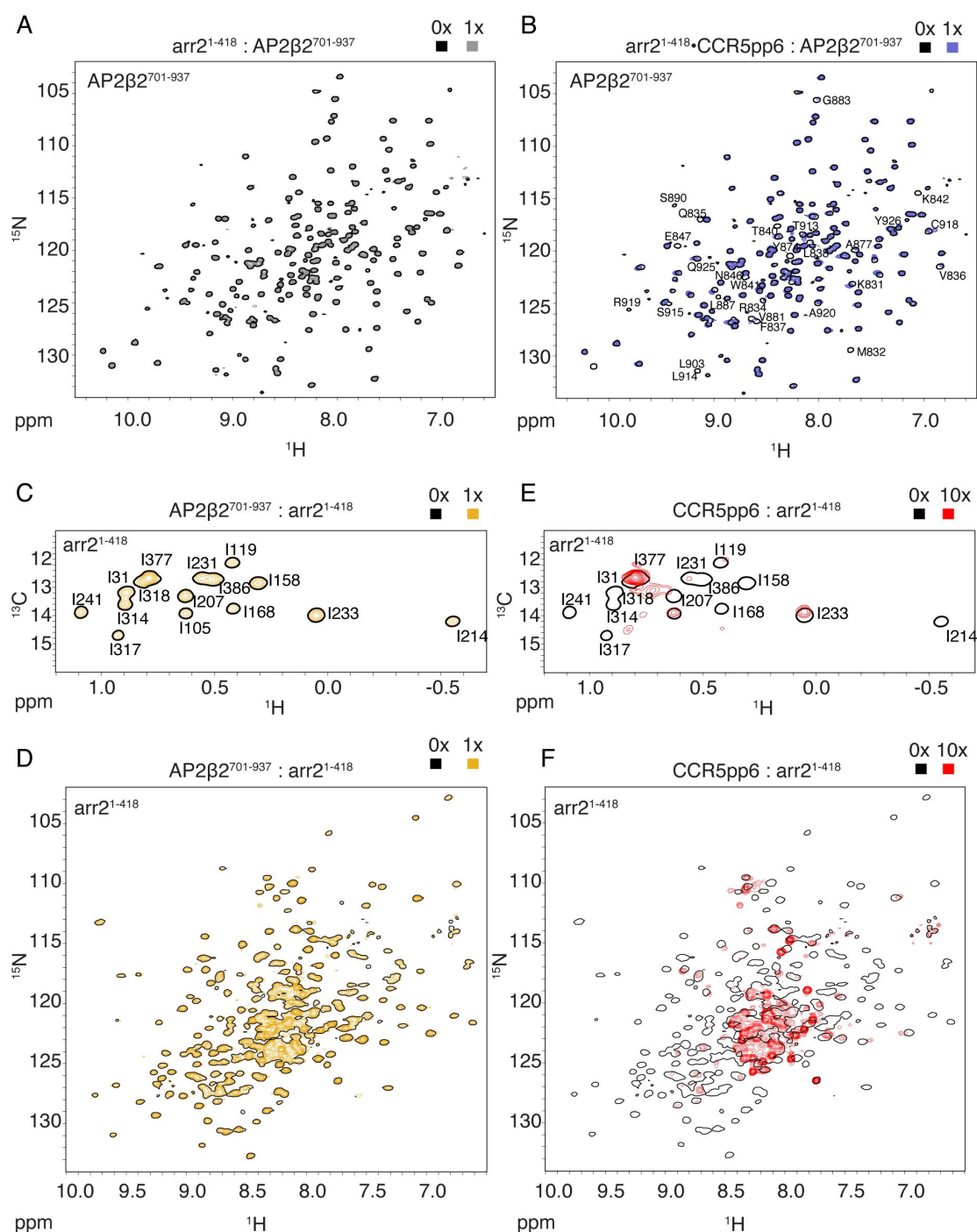


Figure 3 – figure supplement 1.

Arrestin2 interaction with the C-terminal domain of AP2β2 detected by NMR.

(A, B) 1H - ^{15}N TROSY spectra of $AP2\beta2^{701-937}$ in the absence or presence of one molar equivalent of either arrestin2 $^{1-418}$ (A) or arrestin2 $^{1-418}$ •CCR5pp6 (B). Assigned residues undergoing significant intensity reduction are indicated on the spectra. (C, D) Test of interaction of arrestin2 $^{1-418}$ with $AP2\beta2^{701-937}$. (C) 1H - ^{13}C HMQC (Ile-δ1- $^{13}CH_3$, 2H -labeled arrestin2 $^{1-418}$) and (D) 1H - ^{15}N TROSY (^{15}N , 2H -labeled arrestin2 $^{1-418}$) spectra of arrestin2 $^{1-418}$ in the absence or presence of one molar equivalent of $AP2\beta2^{701-937}$. (E, F) Test of interaction of arrestin2 $^{1-418}$ with CCR5pp6. (E) 1H - ^{13}C HMQC (Ile-δ1- $^{13}CH_3$, 2H -labeled arrestin2 $^{1-418}$) and (F) 1H - ^{15}N TROSY (^{15}N , 2H -labeled arrestin2 $^{1-418}$) spectra of arrestin2 $^{1-418}$ in the absence or presence of a ten-molar equivalent of CCR5pp6.

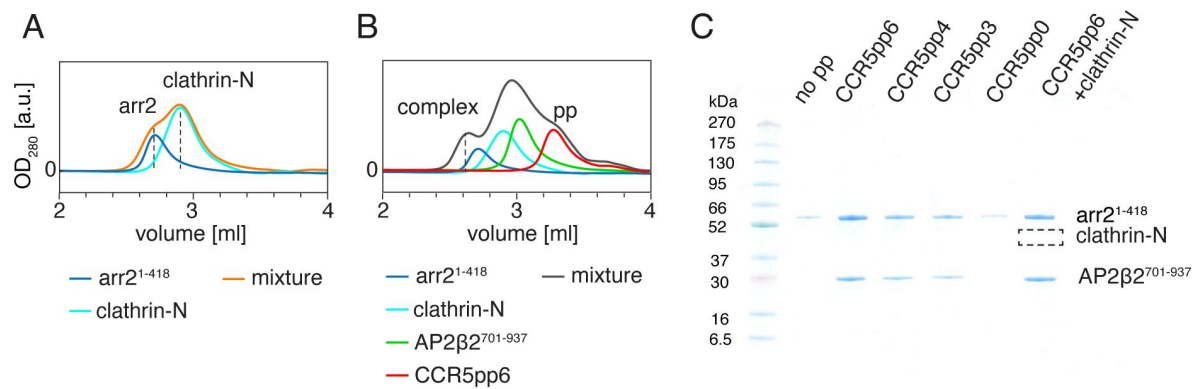


Figure 3-figure supplement 2.

Arrestin2 interaction with clathrin-N and AP2β2 monitored by SEC.

(A) SEC profiles of *arr2*¹⁻⁴¹⁸, *clathrin-N* and their mixture. (B) as in (A) but in the presence of *AP2β2*⁷⁰¹⁻⁹³⁷ and CCR5pp6. (C) SDS-PAGE of sample from the SEC complex fraction (taken at 2.6 ml) monitoring the formation of the *arr2*¹⁻⁴¹⁸/*AP2β2*⁷⁰¹⁻⁹³⁷ complex for various conditions indicated on the top. The empty square indicates the expected position of *clathrin-N*.

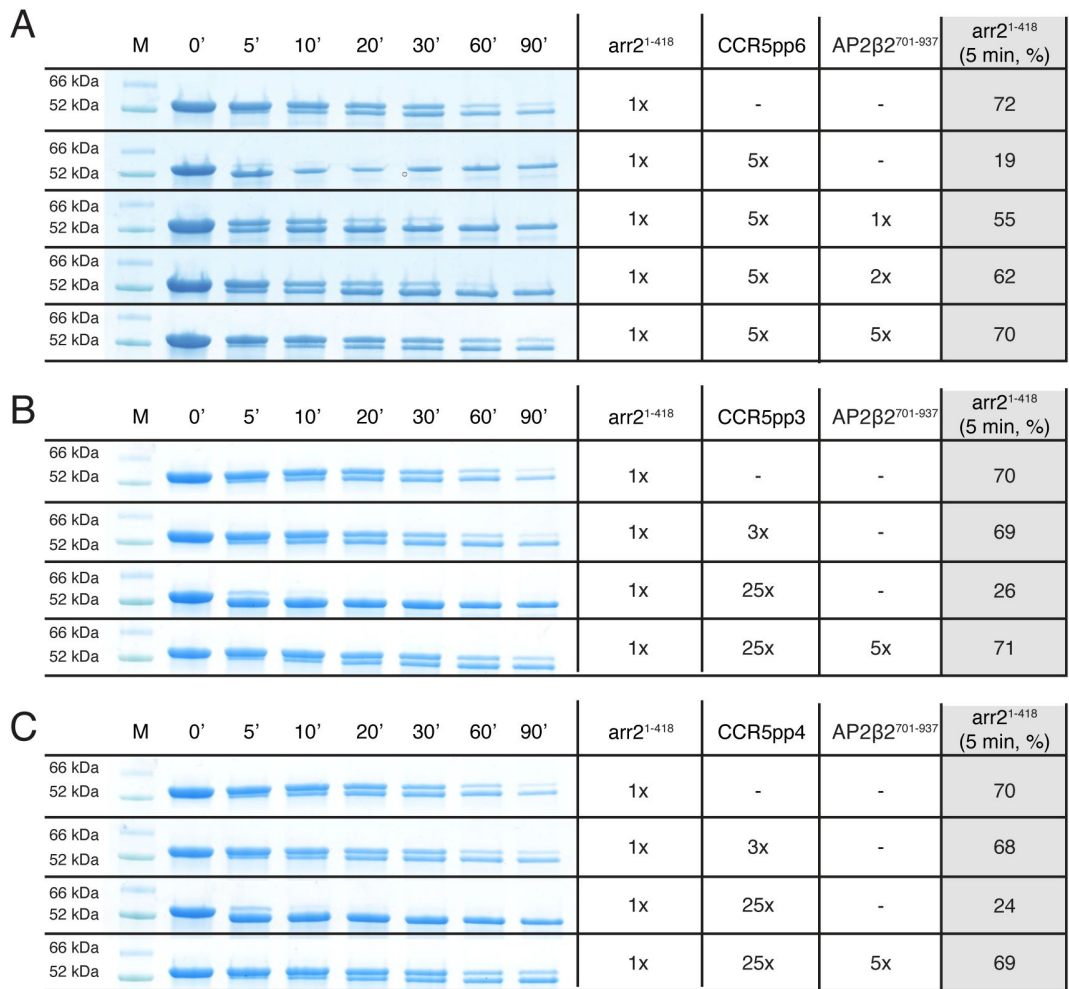


Figure 3-figure supplement 3.

Arrestin2¹⁻⁴¹⁸ trypsin proteolysis assay.

Trypsin proteolysis of arrestin2¹⁻⁴¹⁸ in apo form and in complexes with (A) CCR5pp6, (B) CCR5pp3 and (C) CCR5pp4 and AP2β2⁷⁰¹⁻⁹³⁷ detected by SDS-PAGE. The incubation time is indicated on the top. 'M' indicates the protein size marker. The table on the right indicates sample composition and the relative abundance of arrestin2¹⁻⁴¹⁸ after 5 min. incubation (see Materials and Methods). The activation by the CCR5 phosphopeptides accelerates arrestin2¹⁻⁴¹⁸ proteolysis. The addition of AP2β2⁷⁰¹⁻⁹³⁷ reverses this effect.

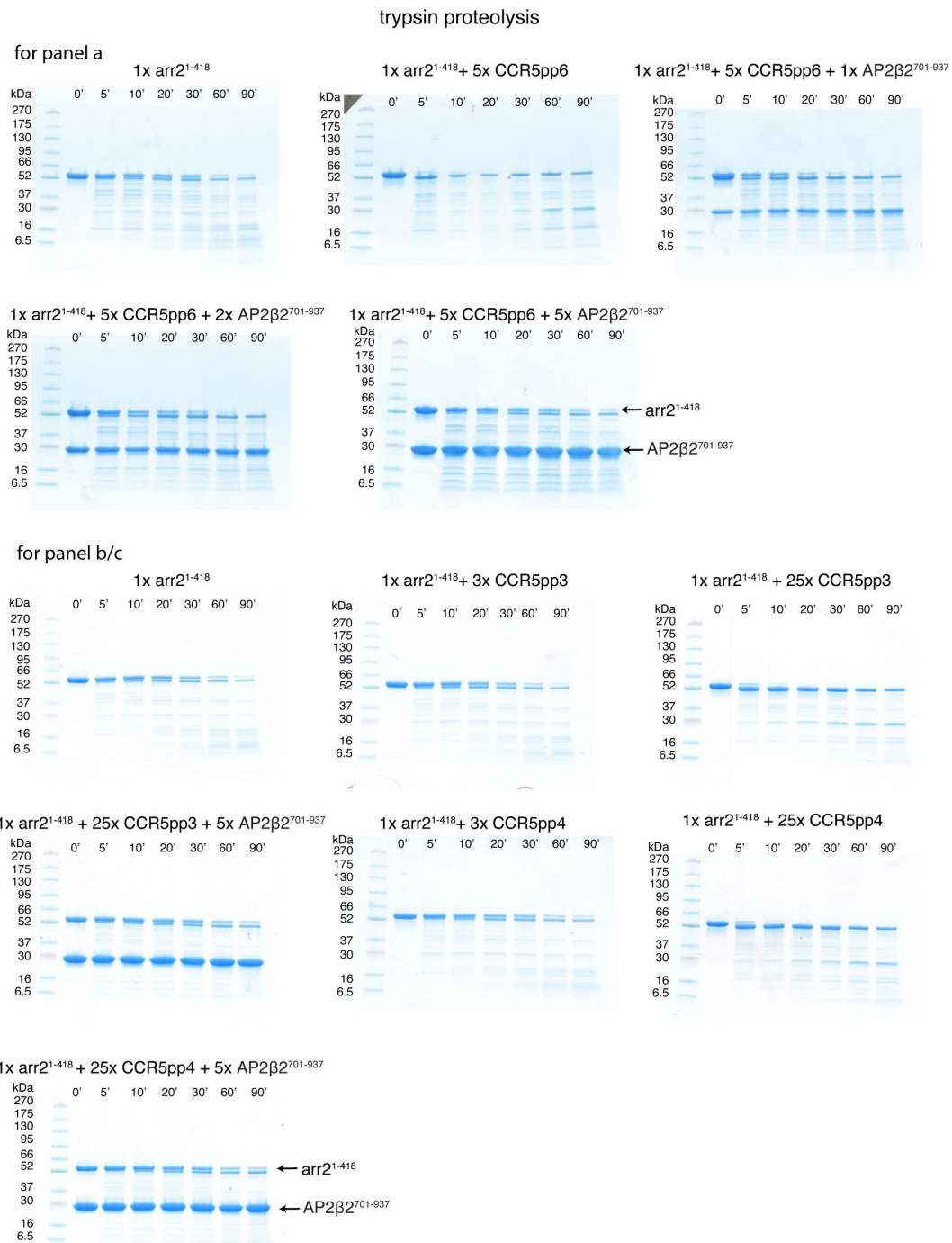


Figure 3–figure supplement 4.

Uncropped SDS-PAGE gels of the arrestin2¹⁻⁴¹⁸ trypsin proteolysis.

SDS-PAGE gels for the data in [Figure 3–figure supplement 3](#). The composition of the sample and the reaction incubation time are indicated at the top. Protein marker bands are indicated on the left. Arrestin2¹⁻⁴¹⁸ runs at ~ 52 kDa, AP2β⁷⁰¹⁻⁹³⁷ at ~ 30 kDa.

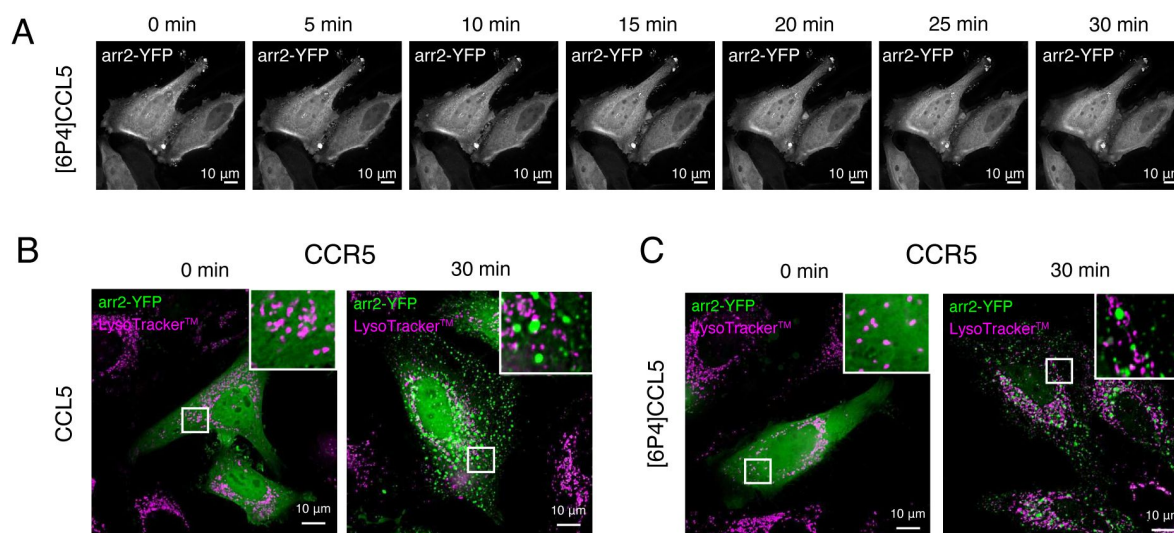


Figure 4-figure supplement 1.

Arrestin2 internalization (control) and re-localization in live HeLa cells upon chemokine stimulation in the presence of CCR5.

(A) HeLa cells transfected only with arrestin2-YFP and stimulated with 6P4[CCL5]. In the absence of the receptor, arrestin2 stays distributed in the cell cytoplasm and nucleus. (B,C) Arrestin2-YFP re-localization in HeLa cells transfected with wild-type CCR5 and arrestin2-YFP genes. Arrestin2-YFP and LysoTracker™ signals were used to monitor the protein trafficking for up to 30 min after CCL5 (B) or 6P4[CCL5] (C) incubation. No overlap between the LysoTracker™ and arrestin2 signal has been detected. White squares indicate zoomed regions of interest.

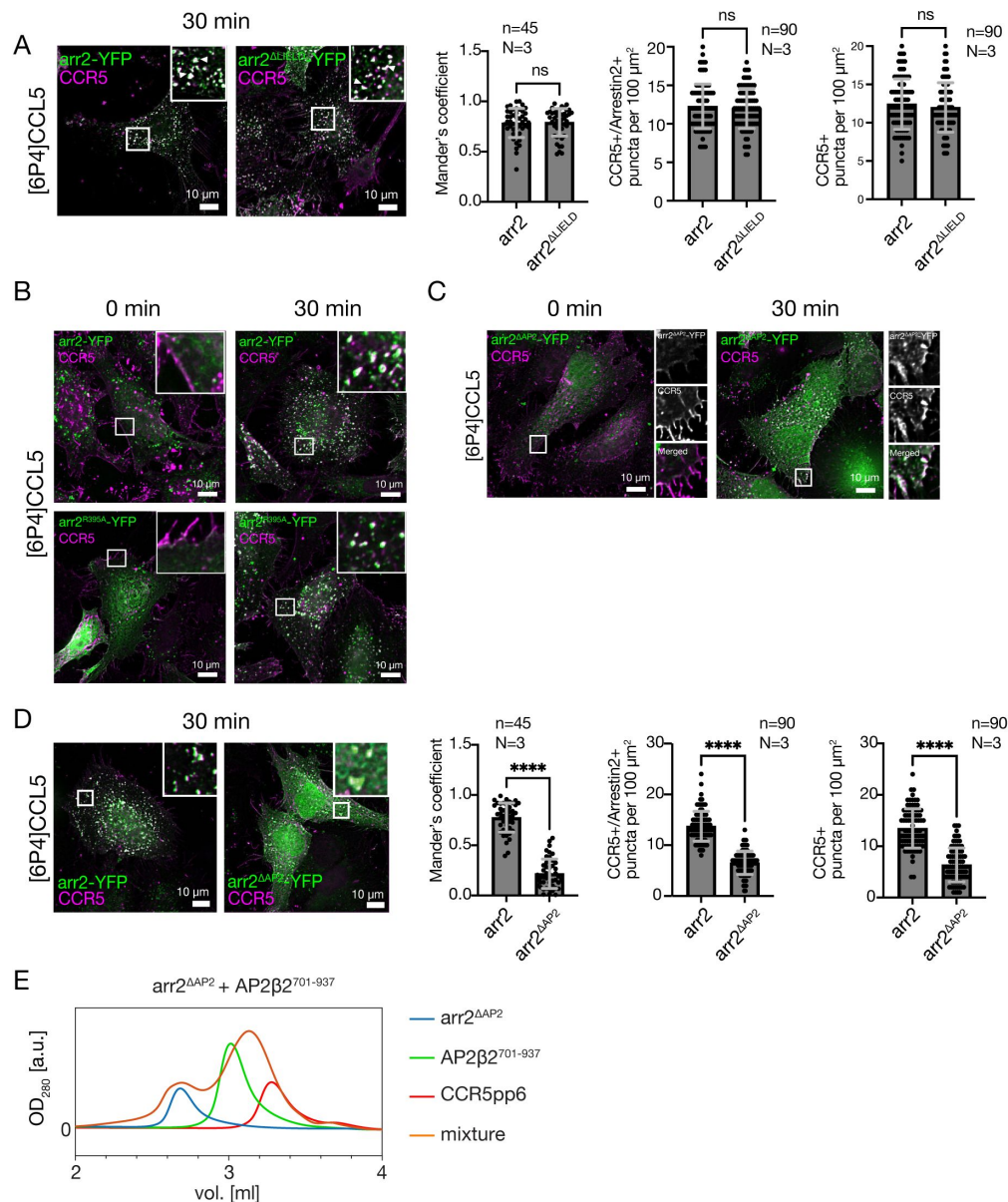


Figure 5-figure supplement 1.

Dependence of CCR5 internalization on the interactions of arrestin2 with clathrin or AP2 monitored in HeLa (CCL2) cells by immunofluorescence microscopy.

(A) CCR5 internalization induced by [6P4]CCL5 in HeLa cells co-transfected with plasmids containing either arrestin2-YFP or arrestin2^{ALIELD}-YFP, which misses the clathrin-binding motif. No significant difference is detected as quantified by the Mander's colocalization coefficient (ns, not significant: $P > 0.9999$) or by counting CCR5⁺/Arrestin2⁺ and only CCR5⁺ puncta 30 min after ligand incubation. Mean and standard deviation are shown for N=3 biological replicates and n=45 ROI from 20 cells (Mander's) or n=90 ROI from 30 cells (puncta counting). (B) CCR5 internalization induced by 6P4[CCL5] in HeLa cells co-transfected with plasmids containing either arrestin2-YFP or arrestin2^{R395A}-YFP, which bears a single mutation in the adaptin-binding motif. The mutation does not affect CCR5 internalization up to 30 min after 6P4[CCL5] stimulation. (C) Recruitment of arrestin2^{ΔAP2}-YFP to CCR5 in the HeLa cell plasma membrane upon [6P4]CCL5 ligand stimulation. (D) CCR5 internalization induced by [6P4]CCL5 in HeLa cells co-transfected with plasmids containing either arrestin2-YFP or arrestin2^{ΔAP2}-YFP. Mean and standard deviation of the Mander's coefficient and the number of puncta were determined as in (A). The absence of the AP2 binding motif in the arrestin2^{ΔAP2}-YFP construct causes a significant (****: $P < 0.0001$) reduction of CCR5 internalization 30 min after ligand incubation. (E) SEC profiles of arrestin2^{ΔAP2}, AP2β2⁷⁰¹⁻⁹³⁷, CCR5pp6 and their mixture. No complex formation is observed.

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Additional information

Data and materials availability

All data needed to evaluate the conclusions in the paper are present in the main text and/or the Supplementary Information.

Author contributions

I.P., S.D., P.I., A.S., and S.G. conceived the study. I.P. and L.A.A generated all the constructs and mutants for the cellular assays. S.D. performed the cellular assays and analyzed the cellular data with guidance from A.S. I.P. expressed and purified proteins. I.P. recorded NMR experiments and analyzed NMR data. I.P. designed and analyzed SEC and trypsin proteolysis assay. I.P. and S.G. wrote the manuscript with inputs from all co-authors.

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Reviewer #1 (Public review):

Petrovic et al. investigate CCR5 endocytosis via arrestin2, with a particular focus on clathrin and AP2 contributions. The study is thorough and methodologically diverse. The NMR titration data clearly demonstrate chemical shift changes at the canonical clathrin-binding site (LIELD), present in both the 2S and 2L arrestin splice variants. To assess the effect of arrestin activation on clathrin binding, the authors compare: truncated arrestin (1-393), full-length arrestin, and 1-393 incubated with CCR5 phosphopeptides. All three bind clathrin comparably, whereas controls show no binding. These findings are consistent with prior crystal structures showing peptide-like binding of the LIELD motif, with disordered flanking regions. The manuscript also evaluates a non-canonical clathrin binding site specific to the 2L splice variant. Though this region has been shown to enhance beta2-adrenergic receptor binding, it appears not to affect CCR5 internalization.

Similar analyses applied to AP2 show a different result. AP2 binding is activation-dependent and influenced by the presence and level of phosphorylation of CCR5-derived phosphopeptides. These findings are reinforced by cellular internalization assays.

In sum, the results highlight splice-variant-dependent effects and phosphorylation-sensitive arrestin-partner interactions. The data argue against a (rapidly disappearing) one-size-fits-all model for GPCR-arrestin signaling and instead support a nuanced, receptor-specific view, with one example summarized effectively in the mechanistic figure.

Weaknesses:

Figure 1 shows regions alphaFold model that are intrinsically disordered without making it clear that this is not an expected stable position. The authors NMR titration data are n=1. Many figure panels require that readers pinch and zoom to see the data.

<https://doi.org/10.7554/eLife.106839.2.sa3>

Reviewer #2 (Public review):

Summary:

Based on extensive live cell assays, SEC, and NMR studies of reconstituted complexes, these authors explore the roles of clathrin and the AP2 protein in facilitating clathrin mediated endocytosis via activated arrestin-2. NMR, SEC, proteolysis, and live cell tracking confirm a strong interaction between AP2 and activated arrestin using a phosphorylated C-terminus of CCR5. At the same time a weak interaction between clathrin and arrestin-2 is observed, irrespective of activation.

These results contrast with previous observations of class A GPCRs and the more direct participation by clathrin. The results are discussed in terms of the importance of short and long phosphorylated bar codes in class A and class B endocytosis.

Strengths:

The 15N,1H and 13C,methyl TROSY NMR and assignments represent a monumental amount of work on arrestin-2, clathrin, and AP2. Weak NMR interactions between arrestin-2 and clathrin are observed irrespective of activation of arrestin. A second interface, proposed by crystallography, was suggested to be a possible crystal artifact. NMR establishes realistic

information on the clathrin and AP2 affinities to activated arrestin with both kD and description of the interfaces.

Weaknesses:

This reviewer has identified only minor weaknesses with the study.

- (1) I don't observe two overlapping spectra of Arrestin2 (1-393) +/- CLTC NTD in Supp Figure 1
- (2) Arrestin-2 1-418 resonances all but disappear with CCR5pp6 addition. Are they recovered with Ap2Beta2 addition and is this what is shown in Supp Fig 2D
- (3) I don't understand how methyl TROSY spectra of arrestin2 with phosphopeptide could look so broadened unless there are sample stability problems?
- (4) At one point the authors added excess fully phosphorylated CCR5 phosphopeptide (CCR5pp6). Does the phosphopeptide rescue resolution of arrestin2 (NH or methyl) to the point where interaction dynamics with clathrin (CLTC NTD) are now more evident on the arrestin2 surface?
- (5) Once phosphopeptide activates arrestin-2 and AP2 binds can phosphopeptide be exchanged off? In this case, would it be possible for the activated arrestin-2 AP2 complex to re-engage a new (phosphorylated) receptor?
- (6) I'd be tempted to move the discussion of class A and class B GPCRs and their presumed differences to the intro and then motivate the paper with specific questions.
- (7) Did the authors ever try SEC measurements of arrestin-2 + AP2beta2+CCR5pp6 with and without PIP2, and with and without clathrin (CLTC NTD? The question becomes what the active complex is and how PIP2 modulates this cascade of complexation events in class B receptors.

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Reviewer #3 (Public review):

Summary:

Overall, this is a well-done study, and the conclusions are largely supported by the data, which will be of interest to the field.

Strengths:

Strengths of this study include experiments with solution NMR that can resolve high-resolution interactions of the highly flexible C-terminal tail of arr2 with clathrin and AP2. Although mainly confirmatory in defining the arr2 CBL 376LIELD380 as the clathrin binding site, the use of the NMR is of high interest (Fig. 1). The ¹⁵N-labeled CLTC-NTD experiment with arr2 titrations reveals a span from 39-108 that mediates an arr2 interaction, which corroborates previous crystal data, but does not reveal a second area in CLTC-NTD that in previous crystal structures was observed to interact with arr2.

SEC and NMR data suggest that full-length arr2 (1-418) binding with 2-adaptin subunit of AP2 is enhanced in the presence of CCR5 phospho-peptides (Fig. 3). The pp6 peptide shows the highest degree of arr2 activation, and 2-adaptin binding, compared to less phosphorylated peptide or not phosphorylated at all. It is interesting that the arr2 interaction with CLTC NTD and pp6 cannot be detected using the SEC approach, further suggesting that clathrin binding is not dependent on arrestin activation. Overall, the data suggest that receptor activation

promotes arrestin binding to AP2, not clathrin, suggesting the AP2 interaction is necessary for CCR5 endocytosis.

To validate the solid biophysical data, the authors pursue validation experiments in a HeLa cell model by confocal microscopy. This requires transient transfection of tagged receptor (CCR5-Flag) and arr2 (arr2-YFP). CCR5 displays a "class B"-like behavior in that arr2 is rapidly recruited to the receptor at the plasma membrane upon agonist activation, which forms a stable complex that internalizes onto endosomes (Fig. 4). The data suggest that complex internalization is dependent on AP2 binding not clathrin (Fig. 5).

The addition of the antagonist experiment/data adds rigor to the study.

Overall, this is a solid study that will be of interest to the field.

<https://doi.org/10.7554/eLife.106839.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Petrovic et al. investigate CCR5 endocytosis via arrestin2, with a particular focus on clathrin and AP2 contributions. The study is thorough and methodologically diverse. The NMR titration data are particularly compelling, clearly demonstrating chemical shift changes at the canonical clathrin-binding site (LIELD), present in both the 2S and 2L arrestin splice variants.

To assess the effect of arrestin activation on clathrin binding, the authors compare: truncated arrestin (1-393), full-length arrestin, and 1-393 incubated with CCR5 phosphopeptides. All three bind clathrin comparably, whereas controls show no binding. These findings are consistent with prior crystal structures showing peptide-like binding of the LIELD motif, with disordered flanking regions. The manuscript also evaluates a non-canonical clathrin binding site specific to the 2L splice variant. Though this region has been shown to enhance beta2-adrenergic receptor binding, it appears not to affect CCR5 internalization.

Similar analyses applied to AP2 show a different result. AP2 binding is activation-dependent and influenced by the presence and level of phosphorylation of CCR5-derived phosphopeptides. These findings are reinforced by cellular internalization assays.

In sum, the results highlight splice-variant-dependent effects and phosphorylation-sensitive arrestin-partner interactions. The data argue against a (rapidly disappearing) one-size-fits-all model for GPCR-arrestin signaling and instead support a nuanced, receptor-specific view, with one example summarized effectively in the mechanistic figure.

We thank the referee for this positive assessment of our manuscript. Indeed, by stepping away from the common receptor models for understanding internalization (b2AR and V2R), we revealed the phosphorylation level of the receptor as a key factor in driving the sequestration of the receptor from the plasma membrane. We hope that the proposed mechanistic model will aid further studies to obtain an even more detailed understanding of forces driving receptor internalization.

Reviewer #2 (Public review):

Summary:

Based on extensive live cell assays, SEC, and NMR studies of reconstituted complexes, these authors explore the roles of clathrin and the AP2 protein in facilitating clathrin-mediated endocytosis via activated arrestin-2. NMR, SEC, proteolysis, and live cell tracking confirm a strong interaction between AP2 and activated arrestin using a phosphorylated C-terminus of CCR5. At the same time, a weak interaction between clathrin and arrestin-2 is observed, irrespective of activation.

These results contrast with previous observations of class A GPCRs and the more direct participation by clathrin. The results are discussed in terms of the importance of short and long phosphorylated bar codes in class A and class B endocytosis.

Strengths:

The ^{15}N , ^1H , and ^{13}C , methyl TROSY NMR and assignments represent a monumental amount of work on arrestin-2, clathrin, and AP2. Weak NMR interactions between arrestin-2 and clathrin are observed irrespective of the activation of arrestin. A second interface, proposed by crystallography, was suggested to be a possible crystal artifact. NMR establishes realistic information on the clathrin and AP2 affinities to activated arrestin, with both k_D and description of the interfaces.

We sincerely thank the referee for this encouraging evaluation of our work and appreciate the recognition of the NMR efforts and insights into the arrestin-clathrin-AP2 interactions.

Weaknesses:

This reviewer has identified only minor weaknesses with the study.

(1) Arrestin-2 1-418 resonances all but disappear with CCR5pp6 addition. Are they recovered with AP2b2 addition, and is this what is shown in Supplementary Figure 2D?

We believe the reviewer is referring to Figure 3 - figure supplement 1. In this figure, the panels E and F show resonances of arrestin2¹⁻⁴¹⁸ (apo state shown with black outline) disappear upon the addition of CCR5pp6 (arrestin2¹⁻⁴¹⁸•CCR5pp6 complex spectrum in red). The panels C and D show resonances of arrestin2¹⁻⁴¹⁸ (apo state shown with black outline), which remain unchanged upon addition of AP2b2⁷⁰¹⁻⁹³⁷ (orange), indicating no complex formation. We also recorded a spectrum of the arrestin2¹⁻⁴¹⁸•CCR5pp6 complex under addition of AP2b2⁷⁰¹⁻⁹³⁷ (not shown), but the arrestin2 resonances in the arrestin2¹⁻⁴¹⁸•CCR5pp6 complex were already too broad for further analysis. This had been already explained in the text.

“In agreement with the AP2b2 NMR observations, no interaction was observed in the arrestin2 methyl and backbone NMR spectra upon addition of AP2b2 in the absence of phosphopeptide (Figure 3-figure supplement 1C, D). However, the significant line broadening of the arrestin2 resonances upon phosphopeptide addition (Figure 3-figure supplement 1E, F) precluded a meaningful assessment of the effect of the AP2b2 addition on arrestin2 in the presence of phosphopeptide”.

(2) I don't understand how methyl TROSY spectra of arrestin2 with phosphopeptide could look so broadened unless there are sample stability problems.

We thank the referee for this comment. We would like to clarify that in general a broadened spectrum beyond what is expected from the rotational correlation time does not necessarily correlate with sample stability problems. It is rather evidence of conformational intermediate exchange on the micro- to millisecond time scale.

The displayed ^1H - ^{15}N spectra of apo arrestin2 already suffer from line broadening due to such intrinsic mobility of the protein. These spectra were recorded with acquisition times of 50 ms (^{15}N) and 55 ms (^1H) and resolution-enhanced by a 60°-shifted sine-bell filter for ^{15}N and a 60°-shifted squared sine-bell filter for ^1H , respectively, which leads to the observed resolution with still reasonable sensitivity. The ^1H - ^{15}N resonances in Fig. 1b (arrestin2¹⁻³⁹³) look particularly narrow. However, this region contains a large number of flexible residues. The full spectrum, e.g. Figure 1-figure supplement 2, shows the entire situation with a clear variation of linewidths and intensities. The linewidth variation becomes stronger when omitting the resolution enhancement filters.

The addition of the CCR5pp6 phosphopeptide does not change protein stability, which we assessed by measuring the melting temperature of arrestin2¹⁻⁴¹⁸ and arrestin2¹⁻⁴¹⁸•CCR5pp6 complex ($T_m = 57^\circ\text{C}$ in both cases). We believe that the explanation for the increased broadening of the arrestin2 resonances is that addition of the CCR5pp6, possibly due to the release of the arrestin2 strand b20, amplifies the mentioned intermediate timescale protein dynamics. This results in the disappearance of arrestin2 resonances.

We have now included the assessment of arrestin2¹⁻⁴¹⁸ and arrestin2¹⁻⁴¹⁸•CCR5pp6 stability in the manuscript:

“The observed line broadening of arrestin2 in the presence of phosphopeptide must be a result of increased protein motions and is not caused by a decrease in protein stability, since the melting temperature of arrestin2 in the absence and presence of phosphopeptide are identical ($56.9 \pm 0.1^\circ\text{C}$)”.

(3) At one point, the authors added an excess fully phosphorylated CCR5 phosphopeptide (CCR5pp6). Does the phosphopeptide rescue resolution of arrestin2 (NH or methyl) to the point where interaction dynamics with clathrin (CLTC NTD) are now more evident on the arrestin2 surface?

Unfortunately, when we titrate arrestin2 with CCR5pp6 (please see Isaikina & Petrovic et. al, Mol. Cell, 2023 for more details), the arrestin2 resonances undergo fast-to-intermediate exchange upon binding. In the presence of phosphopeptide excess, very few resonances remain, the majority of which are in the disordered region, including resonances from the clathrin-binding loop. Due to the peak overlap, we could not unambiguously assign arrestin2 resonances in the bound state, which precluded our assessment of the arrestin2-clathrin interaction in the presence of phosphopeptide. We have made this now clearer in the paragraph ‘The arrestin2-clathrin interaction is independent of arrestin2 activation’

“Due to significant line broadening and peak overlap of the arrestin2 resonances upon phosphopeptide addition, the influence of arrestin activation on the clathrin interaction could not be detected on either backbone or methyl resonances”.

(4) Once phosphopeptide activates arrestin-2 and AP2 binds, can phosphopeptide be exchanged off? In this case, would it be possible for the activated arrestin-2 AP2 complex to re-engage a new (phosphorylated) receptor?

This would be an interesting mechanism. In principle, this should be possible as long as the other (phosphorylated) receptor outcompetes the initial phosphopeptide with higher affinity

towards the binding site. However, we do not have experiments to assess this process directly. Therefore, we rather wish not to further speculate.

(5) Did the authors ever try SEC measurements of arrestin-2 + AP2beta2+CCR5pp6 with and without PIP2, and with and without clathrin (CLTC NTD? The question becomes what the active complex is and how PIP2 modulates this cascade of complexation events in class B receptors.

We thank the referee for this question. Indeed, we tested whether PIP2 can stabilize the arrestin2•CCR5pp6•AP2 complex by SEC experiments. Unfortunately, the addition of PIP2 increased the formation of arrestin2 dimers and higher oligomers, presumably due to the presence of additional charges. The resolution of SEC experiments was not sufficient to distinguish arrestin2 in oligomeric form or in arrestin2•CCR5pp6•AP2 complex. We now mention this in the text:

“We also attempted to stabilize the arrestin2-AP2b2-phosphopeptide complex through the addition of PIP2, which can stabilize arrestin complexes with the receptor (Janetzko et al., 2022). The addition of PIP2 increased the formation of arrestin2 dimers and higher oligomers, presumably due to the presence of additional charges. Unfortunately, the resolution of the SEC experiments was not sufficient to separate the arrestin2 oligomers from complexes with AP2b2”.

Reviewer #3 (Public review):

Summary:

Overall, this is a well-done study, and the conclusions are largely supported by the data, which will be of interest to the field.

Strengths:

(1) The strengths of this study include experiments with solution NMR that can resolve high-resolution interactions of the highly flexible C-terminal tail of arr2 with clathrin and AP2. Although mainly confirmatory in defining the arr2 CBL 376LIED380 as the clathrin binding site, the use of the NMR is of high interest (Figure 1). The 15N-labeled CLTC-NTD experiment with arr2 titrations reveals a span from 39-108 that mediates an arr2 interaction, which corroborates previous crystal data, but does not reveal a second area in CLTC-NTD that in previous crystal structures was observed to interact with arr2.

(2) SEC and NMR data suggest that full-length arr2 (1-418) binding with the 2-adaptin subunit of AP2 is enhanced in the presence of CCR5 phospho-peptides (Figure 3). The pp6 peptide shows the highest degree of arr2 activation and 2-adaptin binding, compared to less phosphorylated peptides or not phosphorylated at all. It is interesting that the arr2 interaction with CLTC NTD and pp6 cannot be detected using the SEC approach, further suggesting that clathrin binding is not dependent on arrestin activation. Overall, the data suggest that receptor activation promotes arrestin binding to AP2, not clathrin, suggesting the AP2 interaction is necessary for CCR5 endocytosis.

(3) To validate the solid biophysical data, the authors pursue validation experiments in a HeLa cell model by confocal microscopy. This requires transient transfection of tagged receptor (CCR5-Flag) and arr2 (arr2-YFP). CCR5 displays a "class B"-like behavior in that arr2 is rapidly recruited to the receptor at the plasma membrane upon agonist activation, which forms a stable complex that internalizes into endosomes (Figure 4). The data suggest that complex internalization is dependent on AP2 binding, not clathrin (Figure 5).

We thank the referee for the careful and encouraging evaluation of our work. We appreciate the recognition of the solidity of our data and the support for our conclusions regarding the distinct roles of AP2 and clathrin in arrestin-mediated receptor internalization.

Weaknesses:

The interaction of truncated arr2 (1-393) was not impacted by CCR5 phospho-peptide pp6, suggesting the interaction with clathrin is not dependent on arrestin activation (Figure 2). This raises some questions.

We thank the referee for raising this concern, as we were also surprised by the discovery that the interaction does not depend on arrestin activation. However, the NMR data clearly show at atomic resolution that arrestin activation does not influence the interaction with clathrin in vitro. Evolutionary, the arrestin-clathrin interaction appears not to be conserved as the visual arrestin completely lacks a clathrin-binding motif. For that reason, we believe that the weak arrestin-clathrin interaction provides more of a supportive role during the internalization rather than the regulatory interaction with AP2, which requires and quantitatively depends on the arrestin2 activation. We have reflected on this in the Discussion:

“Although the generalization of this mechanism from CCR5 to other arr-class B receptors has to be explored further, it is indirectly corroborated in the visual rhodopsin-arrestin1 system. The arr-class B receptor rhodopsin (Isaikina et al., 2023) also undergoes CME (Moaven et al., 2013) with arrestin1 harboring the conserved AP2 binding motif, but missing the clathrinbinding motif (Figure 1-figure supplement 1A)”.

Overall, the data are solid, but for added rigor, can these experiments be repeated without tagged receptor and/or arr2? My concern stems from the fact that the stability of the interaction between arr2 and the receptor may be related to the position of the tags.

We thank the referee for this suggestion, which refers to the cellular experiments; the biophysical experiments were carried out without tags. To eliminate the possibility of tags contributing to receptor-arrestin2 binding in the cellular experiments, we also performed the experiments in the presence of CCR5 antagonist [5P12]CCL5 (Figure 4). These data show that in the case of inactive CCR5, arrestin2 is not recruited to CCR5, nor does it form internalization complexes, which would be the case if the tags were increasing the receptorarrestin interaction. In contrast, if the tags were decreasing the interaction, we would not expect such a strong internalization. As indicated below, we have also attempted to perform our cellular experiments using an N-terminally SNAP-tagged CCR5. Unfortunately, this construct did not express in HeLa cells indicating that SNAP-CCR5 was either toxic or degraded.

Reviewing Editor Comments:

Overall, the reviewers did not suggest much by way of additional experiments. They do suggest several aspects of the manuscript that would benefit from further clarification.

Reviewer #1 (Recommendations for the authors):

(1) The distinction between arrestin 2S and arrestin 2L as relates to the canonical and non-canonical clathrin binding sites would benefit from clarification, particularly because the second binding site depends on the splice variant. This is something that some readers may not be familiar with (particularly young ones that are hopefully part of the intended readership).

We thank the referee for this suggestion. We would like to emphasize that in our work, only the long arrestin2 splice variant was used, which contains both binding sites. We have now introduced the splice variants and their relation to the clathrin binding sites in the text.

In section ‘Localizing and quantifying the arrestin2-clathrin interaction by NMR spectroscopy’:

“Clathrin and arrestin interact in their basal state (Goodman et al., 1996), and a structure of a complex between arrestin2 and the clathrin heavy chain N-terminal domain (residues 1-363, named clathrin-N in the following) has been solved by X-ray crystallography (PDB:3GD1) in the absence of an arrestin2-activating phosphopeptide (Kang et al., 2009). This structure (Figure 1-figure supplement 1B) suggests a 2:1 binding model between arrestin2 and clathrinN. The first interaction (site I) is observed between the ³⁷⁶LIELD³⁸⁰ clathrin-binding motif of the arrestin2 CBL and the edge of the first two β-sheet blades of clathrin-N, whereas the second interaction (site II) occurs between arrestin2 residues ³³⁴LLGDLA³³⁹ and the 4th and 5th blade of clathrin-N. The latter arrestin interaction site is not present in the arrestin2 splice variant arrestin2S (for short) where an 8-amino acid insert (residues 334-341) between β-strands 18 and 19 is removed (Kang et al., 2009)”.

Section ‘The arrestin2-clathrin interaction is independent of arrestin2 activation’

“Figure 2A (left) shows the intensity changes (full spectra in Figure 2-figure supplement 1A) of the clathrin-N ¹H-¹⁵N TROSY resonances [assignments transferred from BMRB, ID:25403 (Zhuo et al., 2015)] upon addition of a one-molar equivalent of arrestin2¹⁻³⁹³. A significant intensity reduction due to line broadening is detected for clathrin-N residues 39-40, 48-50, 62-72, 83-90, 101-106, and 108. These residues form a clearly defined binding region at the edges of blade 1 and blade 2 of clathrin-N (Figure 2A, right), which corresponds to interaction site I in the 3GD1 crystal structure, involving the conserved arrestin2 ³⁷⁶LIELD³⁸⁰ motif. However, no significant signal attenuation was observed for clathrin-N residues in blade 4 and blade 5, which would correspond to the crystal interaction site II with arrestin2 residues ³³⁴LLGDLA³³⁹ that are absent in the arrestin2S splice variant. Thus only one arrestin2 binding site in clathrin-N is detected in solution, and site II of the crystal structure may be a result of crystal packing”.

(2) Acronym density is high throughout. While many are standard in the clathrin literature, this could hinder accessibility for readers with a GPCR or arrestin focus.

We agree with the referee. The acronyms were hard to avoid. The most non-obvious acronym seems ‘CLTC-NTD’ for the N-terminal domain of the clathrin heavy chain, which uses the non-obvious, but common gene name CLTC for the clathrin heavy chain. We have now replaced ‘CLTC-NTD’ by ‘clathrin-N’ and hope that this makes the text easier to follow.

(3) The NMR section, while impressive in scope, had writing that was more difficult to follow than the rest. I am curious what percentage of resonance could be assigned.

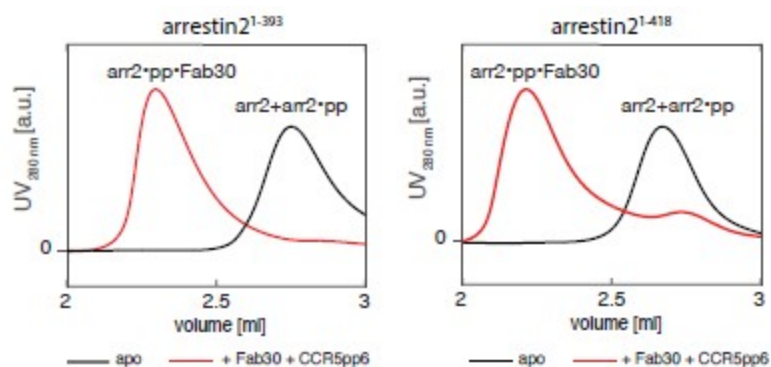
We apologize if the NMR sections of this manuscript were unclear. We attempted to provide a very detailed description of the experimental setup and the spectral results. Being experienced NMR spectroscopists, we have tried very hard to obtain good 3D triple resonance spectra for assignments, but their sensitivity is very low. We believe that this is due to the microsecond dynamics present in the system, which makes the heteronuclear transfers inefficient. So far, we have been able to assign ~30% of the visible arrestin2 resonances. We are still validating the assignments and are working on the analysis and an explanation for this arrestin2 behavior. Therefore, at this point, we want to refrain from stronger statements besides that considerable intrinsic microsecond dynamics is impeding the assignment process.

(4) It may be worth noting in the main text that truncated arrestins have slightly higher basal activation. I was curious why the truncated arrestin was not chosen for the AP2 NMR titrations. Presumably, an effect would be more likely to be seen.

While some truncated arrestin2 variants (comprising residues 1-382 or 1-360) indeed show higher basal activity than the full-length arrestin2, they typically completely lack the b20 strand (residues 386-390), which is crucial for the formation of a parallel b-sheet with strand b1, and whose release governs arrestin activation. Our truncated arrestin2 construct comprises residues 1-393 and contains strand b20. In our experience, no significant difference in basal activity, as assessed by Fab30 binding, was detected for arrestin2¹⁻³⁹³ and arrestin2¹⁻⁴¹⁸ (Author response image 1).

Author response image 1.

SEC profiles showing arrestin2¹⁻³⁹³ (left) and arrestin2¹⁻⁴¹⁸ (right) activation by the CCR5pp6 phosphopeptide as assayed by Fab30 binding. The active ternary arrestin2-phosphopeptide-Fab30 complex elutes at a lower volume than the inactive apo arrestin2 or the binary arrestin2-phosphopeptide complex. Both arrestin2 constructs are activated by the phosphopeptide to a similar level as assessed by the integrated SEC volumes.



We want to emphasize that we used full-length arrestin2¹⁻⁴¹⁸ in order to assess the AP2 interaction, as the crystal structure of arrestin2 peptide-AP2 (PDB:2IV8) shows residues past the residue 393 involved in binding.

PDB codes are currently not accompanied by corresponding literature citations throughout. Please add these.

Thank you for this suggestion. In the manuscript, we were careful to provide the full literature citation the first time each PDB code is mentioned. To avoid redundancy and maintain clarity, we rather do not want to repeat the citations with every subsequent mentioning of the PDB code.

(5) The AlphaFold model could benefit from a more transparent discussion of prediction confidence and caveats. The younger crowd (part of the presumed intended readership) tends to be more certain that computational output is 'true'. Figure 1A shows long loops that are likely regions of low confidence in the prediction. Displaying expected disordered regions as transparent or color-coded would help highlight these as flexible rather than stable, especially for that same younger readership.

We need to explain that the AlphaFold model of arrestin2 was only used to visualize the clathrin-binding loop and the 344-loop of the arrestin2 C-domain, which are not detected in

the available apo bovine (PDB:1G4M) and apo human (PDB:8AS4) arrestin2 crystal structures. However, the AlphaFold model of arrestin2 is basically identical to the crystal structures in the regions that are visible in the crystal structures. We have clarified this now in the caption to Figure 1.

“The model was used to visualize the clathrin-binding loop and the 344-loop of the arrestin2 C-domain, which are not detected in the available crystal structures of apo arrestin2 [bovine: PDB 1G4M (Han et al., 2001), human: PDB 8AS4 (Isaikina et al., 2023)]. In the other structured regions, the model is virtually identical to the crystal structures”.

(6) Several figure panels were difficult to interpret due to their small size. Especially microscopy insets, where I needed to simply trust that the authors were accurately describing the data. Enlarging panels is essential, and this may require separating them into different figures.

We appreciate the referee’s concern regarding figure readability. However, we want to indicate that all our figures are provided as either high-resolution pixel or scalable vector graphics, which allow for zooming in to very fine detail, either electronically or in print. This ensures that microscopy insets and other small panels can be examined clearly when viewed appropriately. We believe the current layout of the figures is necessary to be able to efficiently compare the data between different conditions.

Many figure panels had text size that was too small. Font inconsistencies across figures also stand out.

We apologize for this. We have now enlarged the font size in the figures and made the styles more consistent.

For Fig. 1F, consider adding individual data points and error bars.

Thank you for this suggestion. However, Figure 1F already contains the individual data points, with colored circles corresponding to the titration condition. As we did not have replicates of the titration, no error bars are shown. However, the close agreement of the theoretical fit with the individual measured data points stemming from different experiments shows that the statistical errors are indeed very small. We have estimated an overall error for the K_d (as indicated in panel F, right) by error propagation based on an estimate of the chemical shift error as obtained in the NMR software POKY (based on spectral noise).

Reviewer #2 (Recommendations for the authors):

(1) I don't observe two overlapping spectra of Arrestin2 (1393) +/- CLTC NTD in Supplementary Figure 1.

As explained above all the spectra are shown as scalable vector graphics. The overlapping spectra are visible when zoomed in.

(2) I'd be tempted to move the discussion of class A and class B GPCRs and their presumed differences to the intro and then motivate the paper with specific questions.

We appreciate the referee’s suggestion and had a similar idea previously. However, as we do not have data on other class-A or class-B receptors, we rather don’t want to motivate the entire manuscript by this question.

Reviewer #3 (Recommendations for the authors):

(1) What happens with full-length arr2 (1-418) when the phospho-peptide pp6 is added to the reaction? It's unclear to me that 1-418 would behave the same as 1-393 because the arr2 tail of 1-393 is likely sufficiently mobile to accommodate binding to CLTC NTD. I suggest attempting this experiment for added rigor.

We believe that there is a misunderstanding. The 1-393 and 1-418 constructs differ by the disordered C-terminal tail, which is not involved in the clathrin interaction with the arrestin2 376-380 (LIELD) residues. Accordingly, both 1-393 and 1-418 constructs show almost identical interactions with clathrin (Figure 2A and 2C). Moreover, the phospho-activated arrestin2¹⁻³⁹³ (Figure 2B) interacts identically with clathrin as inactive arrestin2¹⁻³⁹³ and inactive arrestin2¹⁻⁴¹⁸. We believe that this comparison is sufficient for the conclusion that arrestin activation does not play a role in arrestin-clathrin binding.

(2) If the tags were moved to the N-terminus of the receptor and/or arr2, I wonder if the complex is as stable (Figure 4)?

We thank the referee for their suggestion. We have indeed attempted to perform our experiments using an N-terminally SNAP-tagged CCR5. Unfortunately, this construct did not express in the HeLa cells indicating that SNAP-CCR5 was either toxic or degraded. Unfortunately, as the lab is closing due to the retirement of the PI, we are not able to repeat these experiments with further differently positioned tags. We refer also to our answer above that the experiments with the antagonist [5P12]CCL5 present a certain control.

(3) A biochemical assay to measure receptor internalization, in addition to the cell biological approach (Figure 5), would add additional rigor to the study and conclusions.

We tried to measure internalization using a biochemical approach. We tried to pull-down CCR5 from HeLa cells and assess arrestin binding. Unfortunately, even using different buffer conditions, we found that CCR5 was aggregating once solubilized from membranes, preventing us from doing this analysis. We had a similar problem when we exogenously expressed CCR5 in insect cells for purification purposes. We have long experience with CCR5, and this receptor is very aggregation-prone due to extended charged surfaces, which interact with the chemokines.

As an alternative, and in support of the cellular immunofluorescence assays, we also attempted to obtain internalization data via FACS using a CCR5 surface antibody (CD195 Monoclonal Antibody eBioT21/8). CD195 recognizes the N-terminus of the receptor. Unfortunately, the presence of the chemokine ligand (~ 8 kDa) interferes with antibody binding, precluding the quantitative biochemical assessment of the arrestin2 mutants on the CCR5 internalization.

For these reasons, we were particularly careful to quantify CCR5 internalization from the immunofluorescence microscopy data using colocalization coefficients as well as puncta counting (Figure 4+5).

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